Branny Branfield

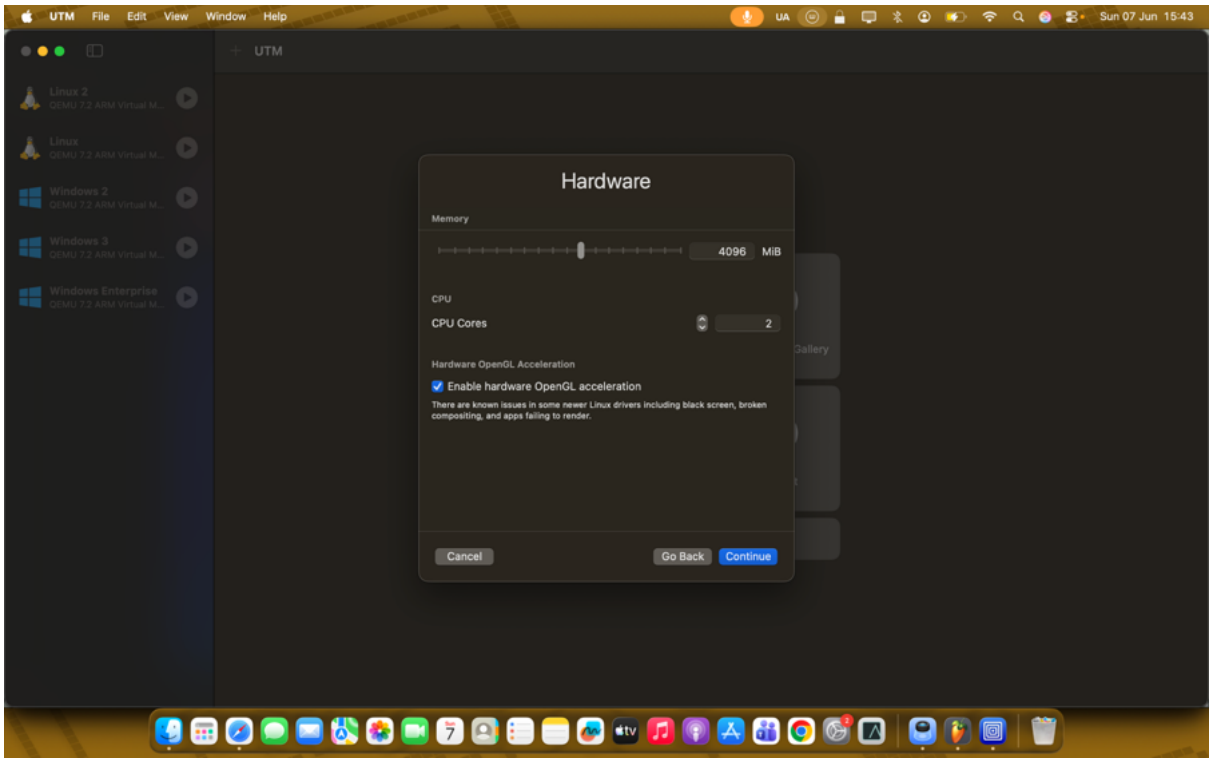
ITLXA0-22

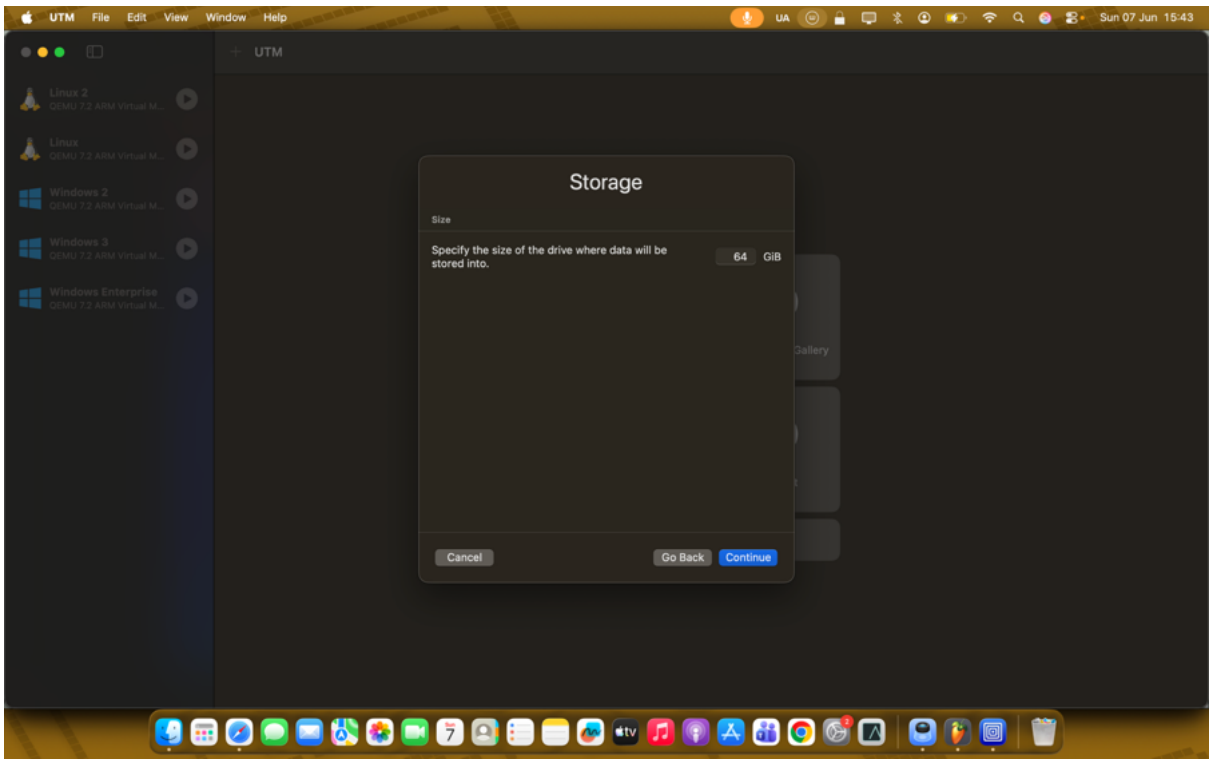
Formative Assessment

Bedfordview Campus: Eduvos

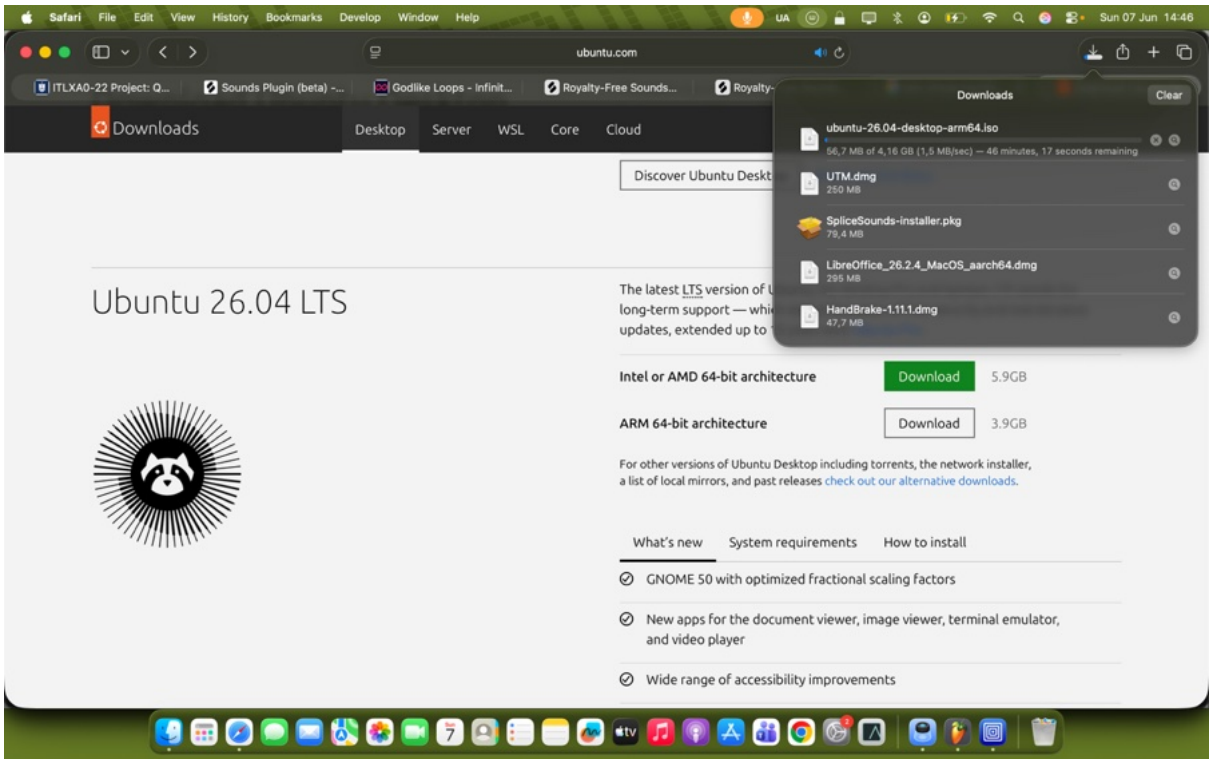
Student Number : EDUV12936584

Question 1 Virtual Machine Installation

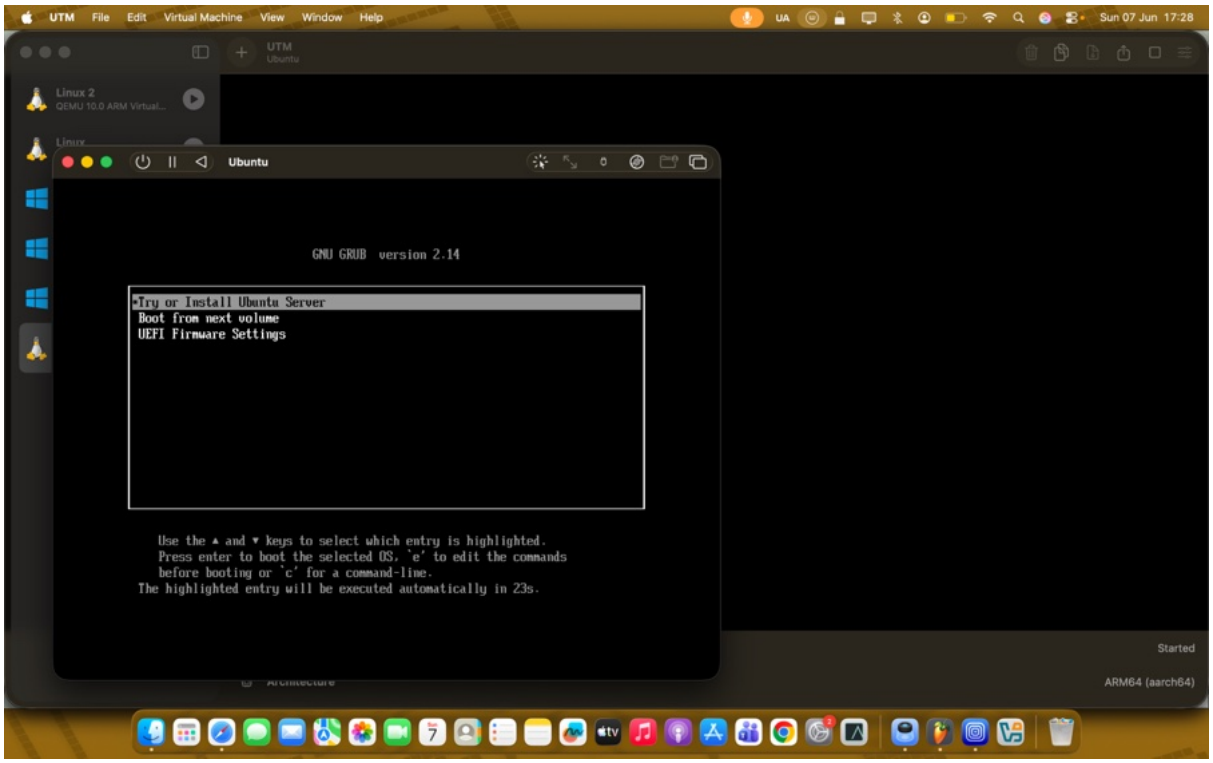


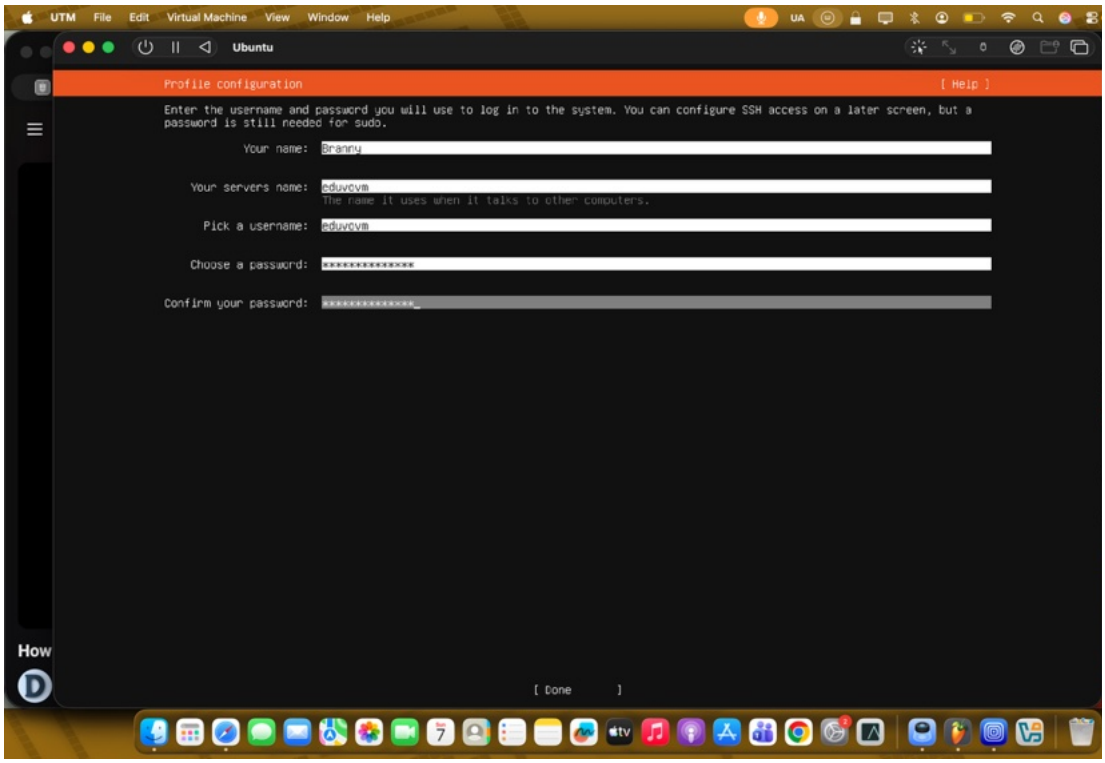


Creating a VM with 4096 RAM, 2 Core CPU AND 64Gib Storage using UTM Virtual Machine for MacOS Operating System with ARM64 Silicon M chip.

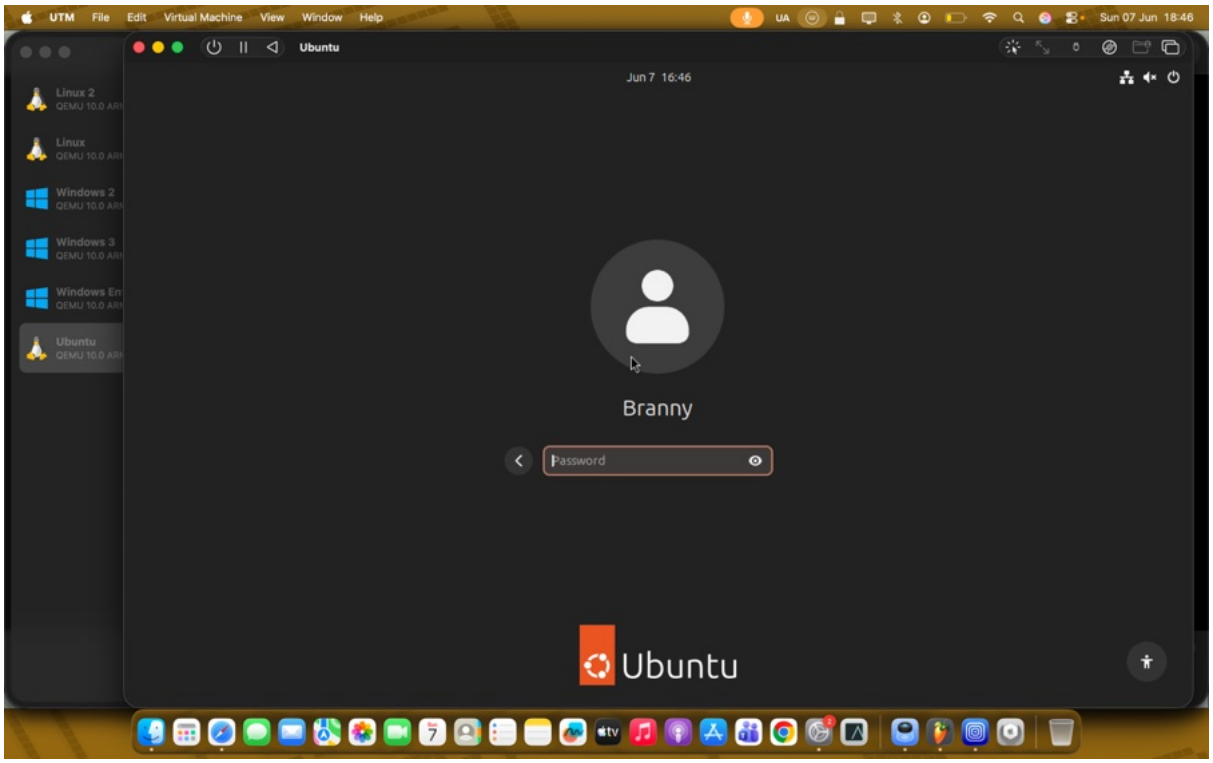


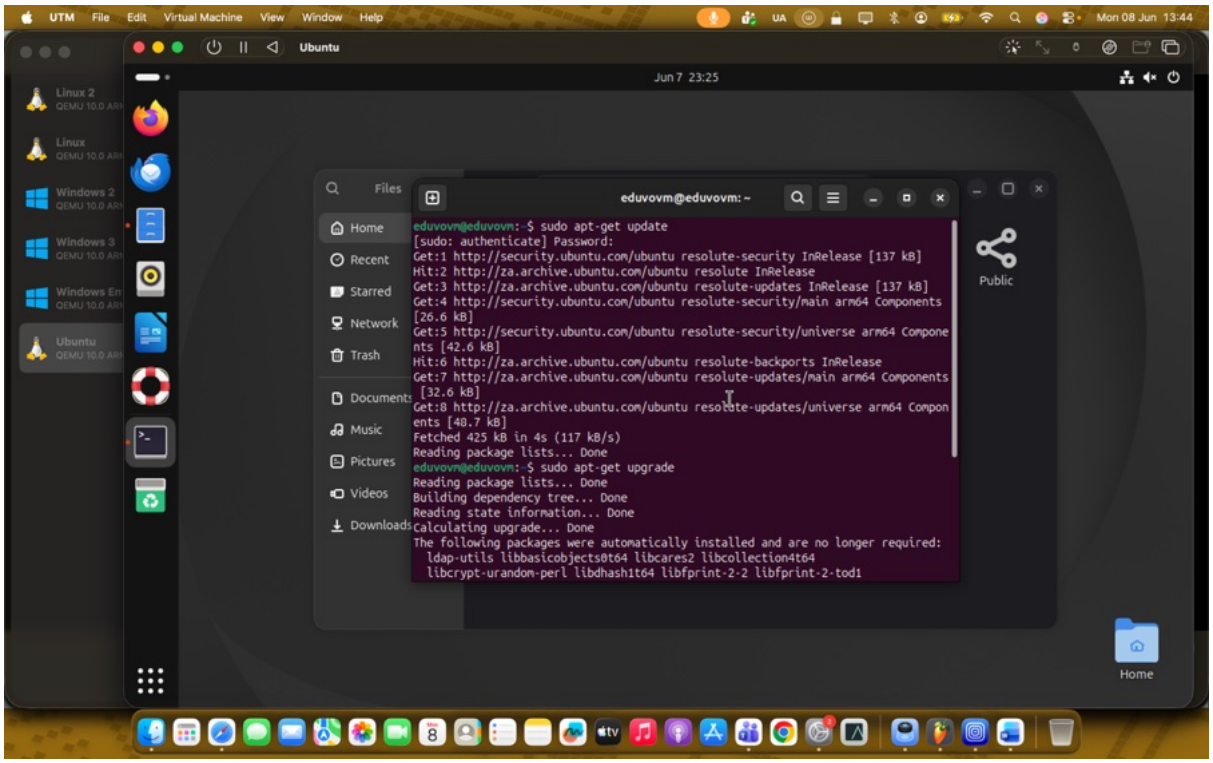
Installing Ubuntu ISO file for Mac ARM64bit Architecture on the official web page of Ubuntu.





Creating a secure user account on Ubuntu using Profile Configuration during the process of setting up a new Virtual Machine.





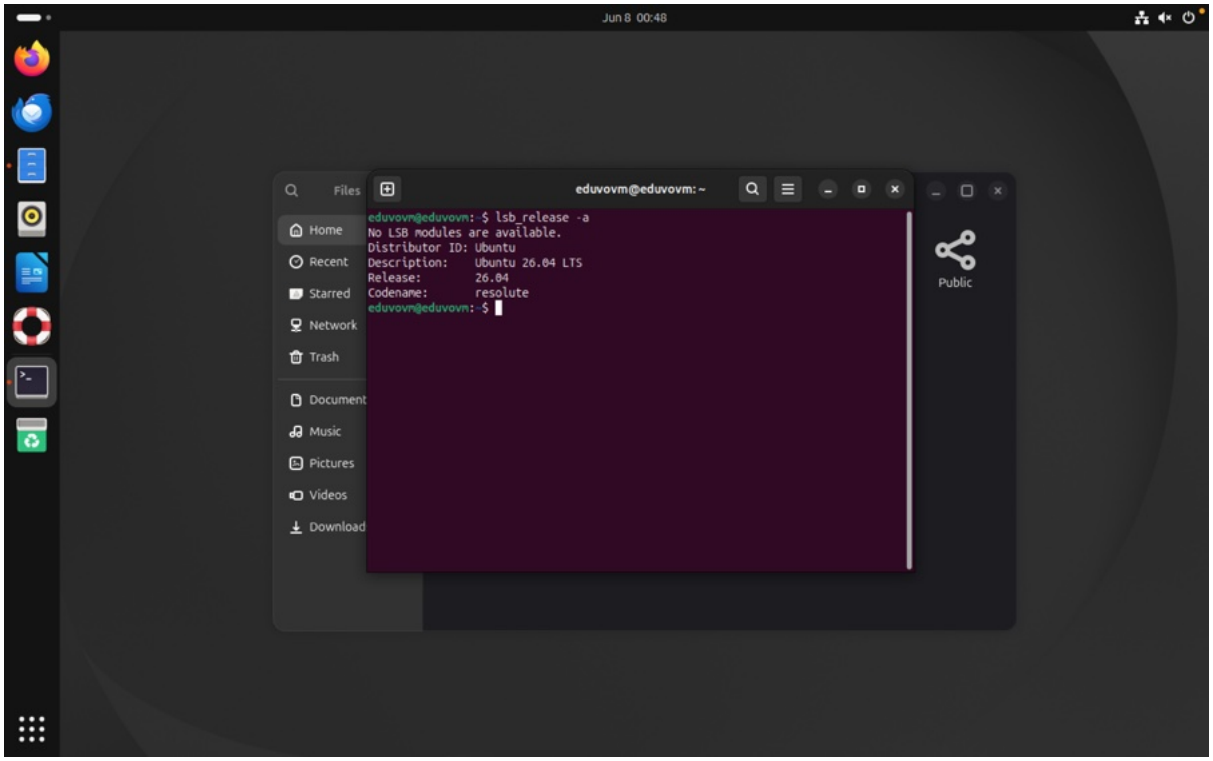
Logging into my new Virtual Machine System account and performing a system update on the terminal using sudo apt-get update/apt-get upgrade.

a) I have chosen a VM configuration of 4096 RAM, 2 Core CPU AND 64Gib Storage to save my physical computers resource while and still be able to perform basic tasks on Ubuntu like running terminal, launching application, modifying System Setting, Creating and managing file, Creating and editing documents as well as web surfing .

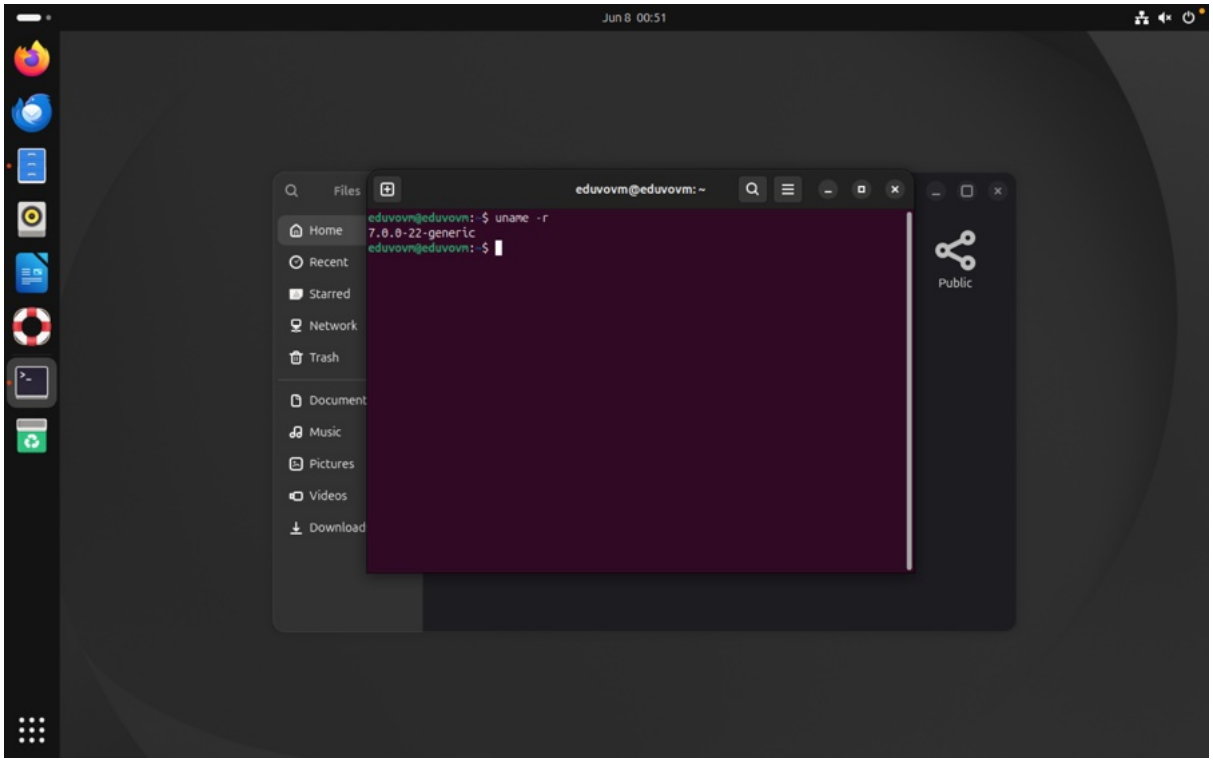
b) Two risks of poor VM configuration include Resource Misallocation in which using a lot of memory(compared to what's available.) could resort into memory swap to disk slowing down the processing speed. And Security risks like VM Escape where a VM could be compromised and exploited by a malicious actor(hacker) to escape VM and gain control on host system.

c) Virtual Machine are Preferred in Training Environments due to the fact of the offering a safe sandboxed environment allowing users to practice on a Operating System without damaging hardware and be able to recover with snapshots after corrupting Operating System.

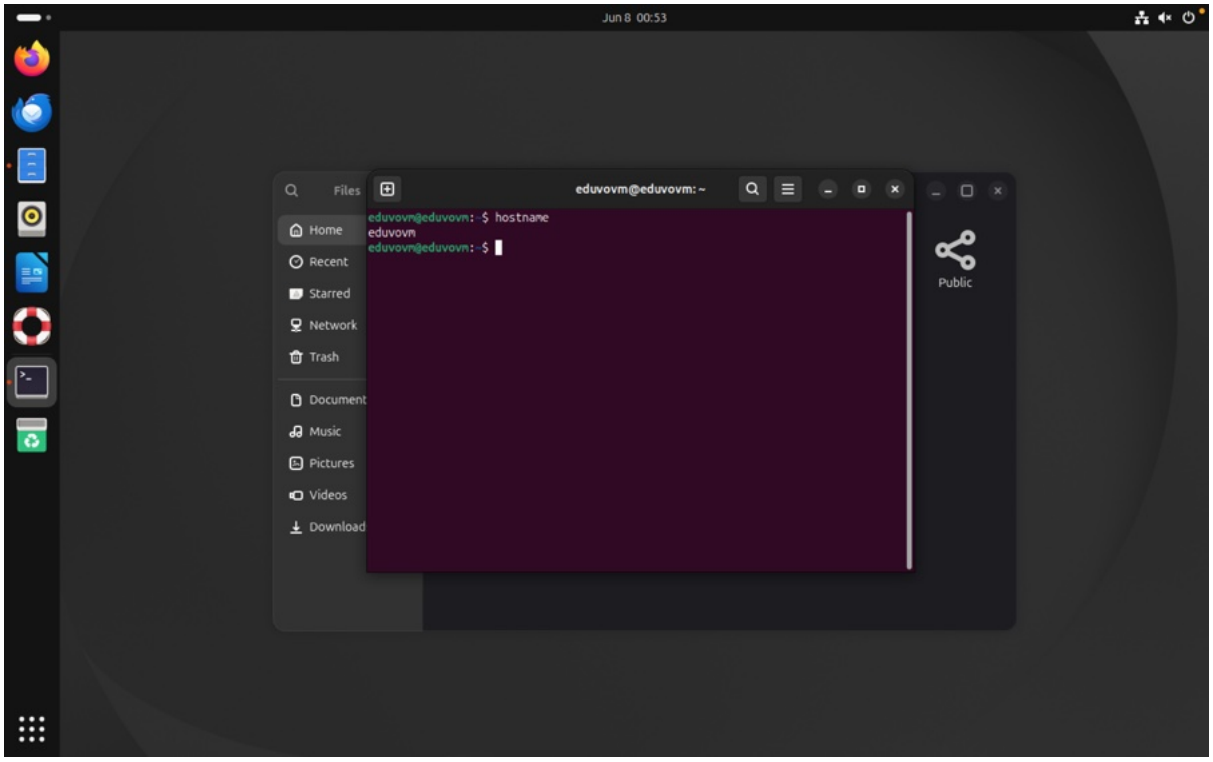
Question 2 System Verification



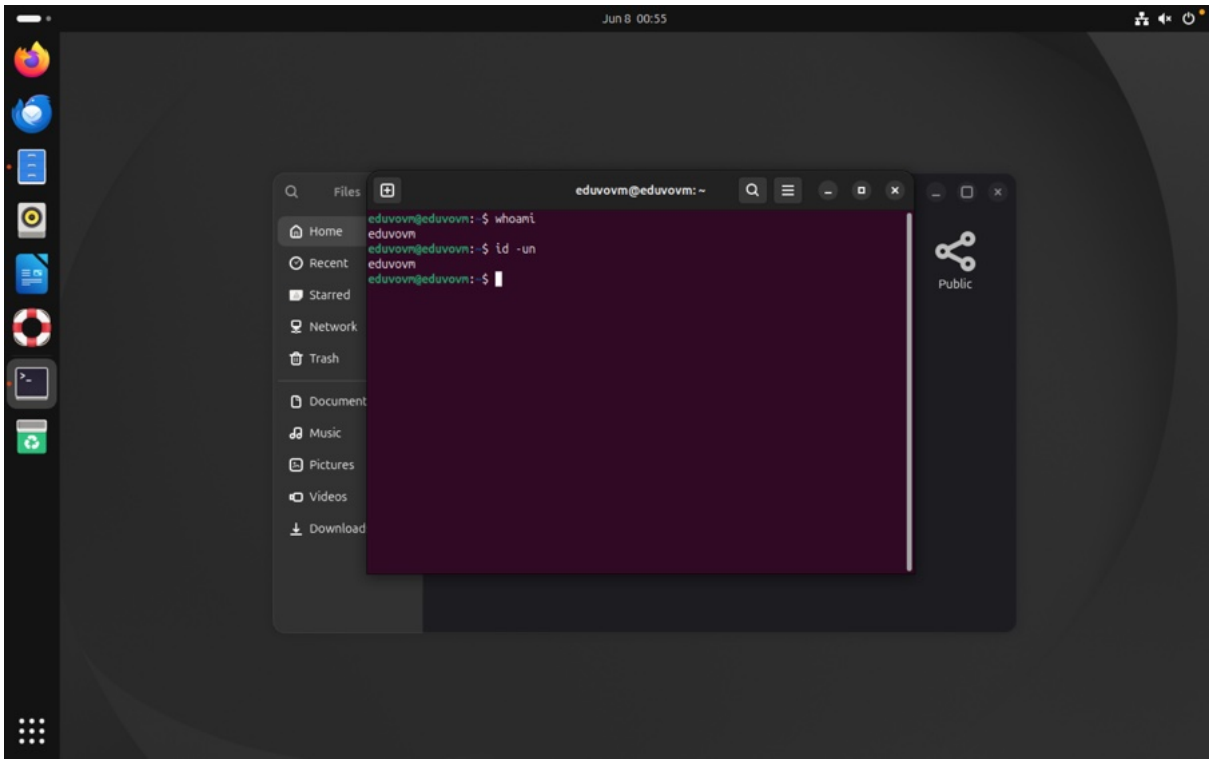
Using command lsb_release -a to display Ubuntu Version which is Ubuntu 26.04 LTS.



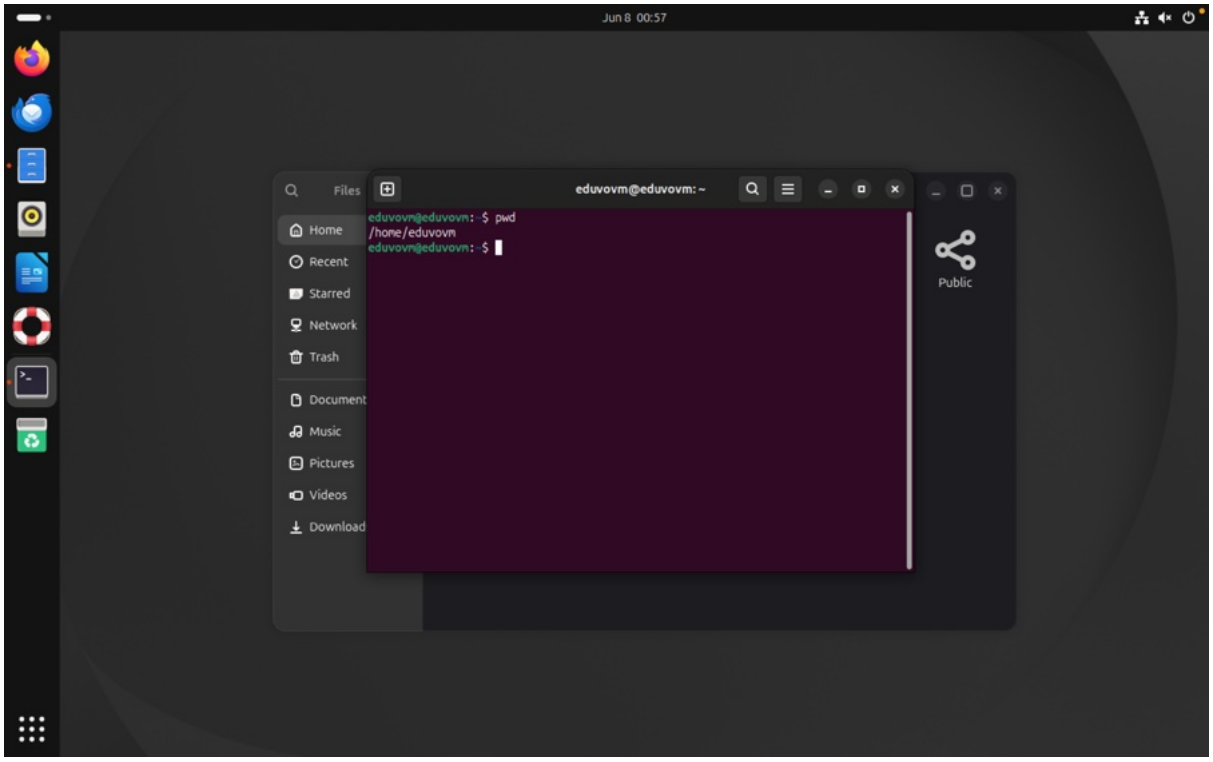
Using command uname -r to display Kernal Version which is 7.0.0-22-generic



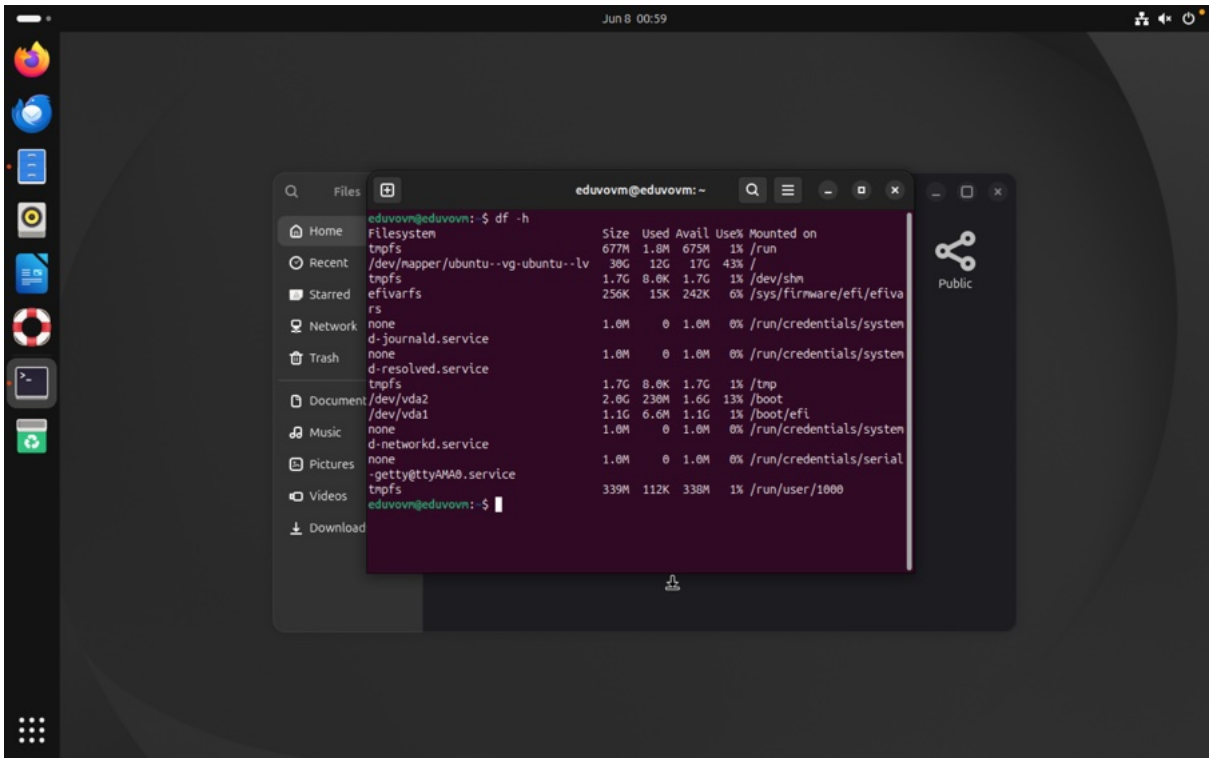
Using command hostname to display Hostname which is eduvovm.



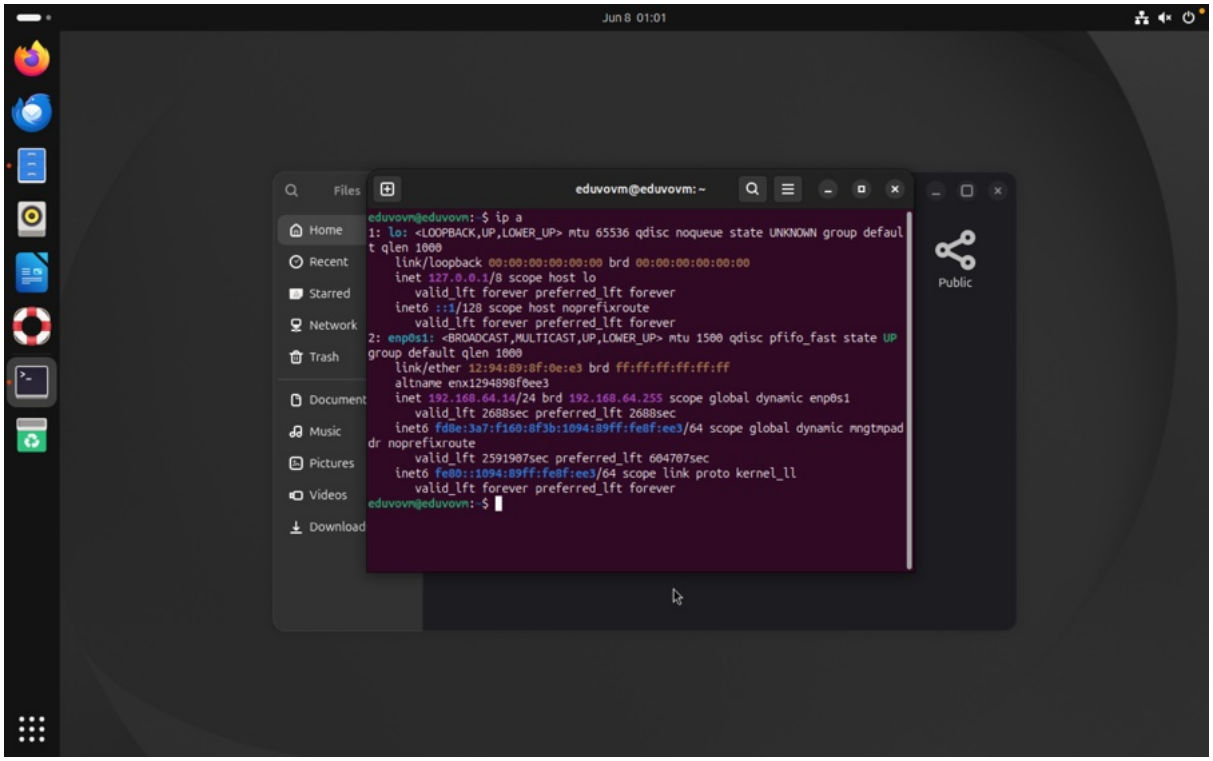
Using command whoami and id -un to display Current User which is eduvovm.



Using command pwd to display current directory which is /home/eduvovm.



Using command df -h to display Disk Usage.



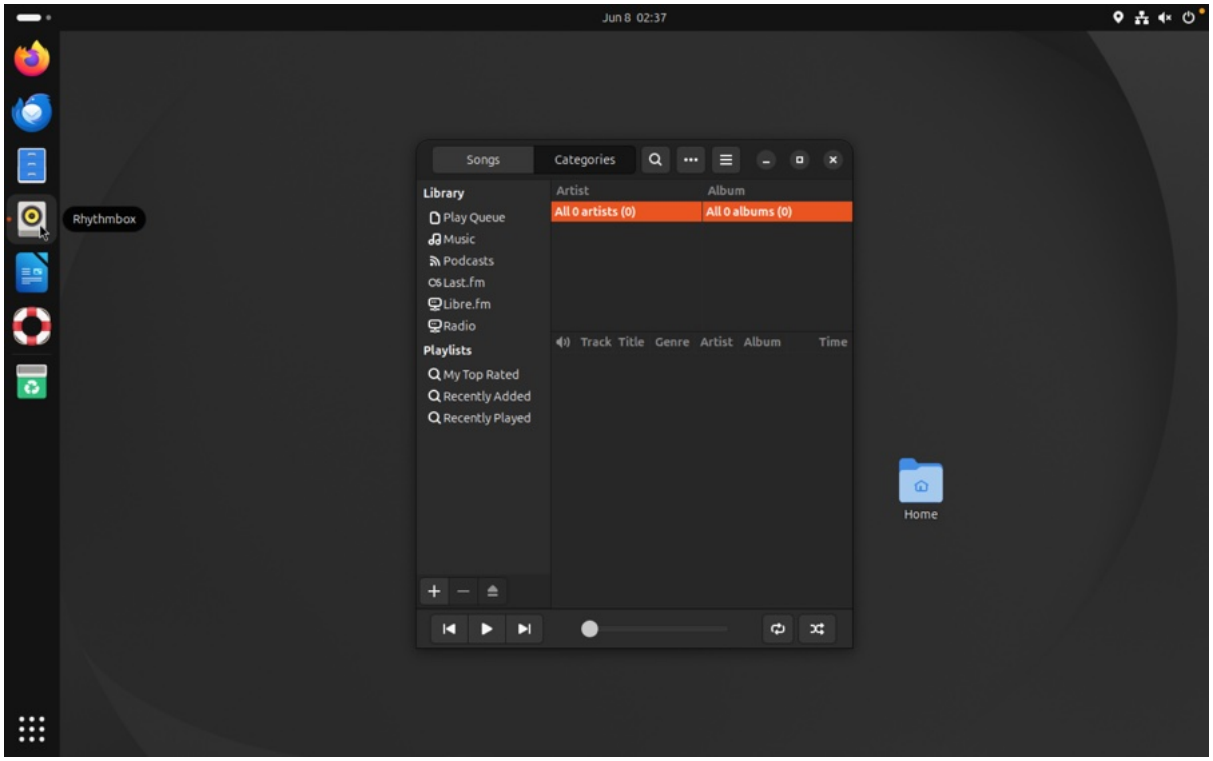
Using command ip a to display Network interface.

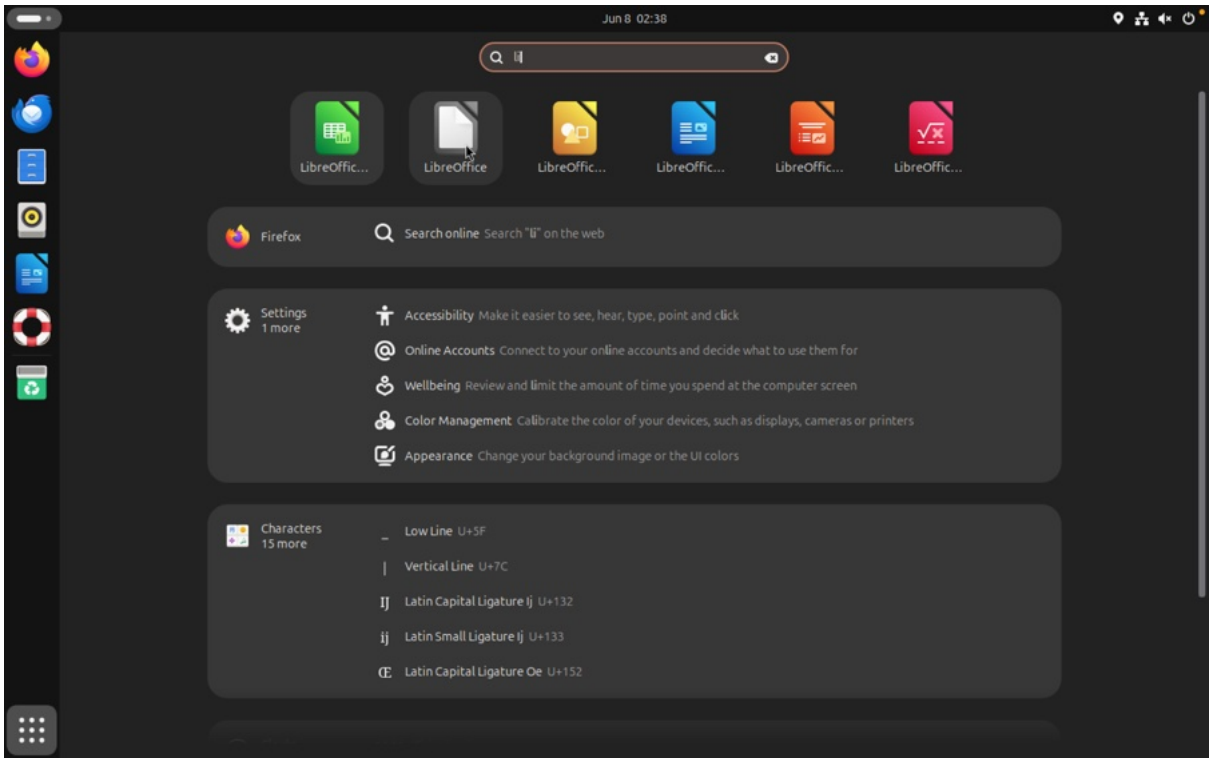
b) How does this information help with troubleshooting

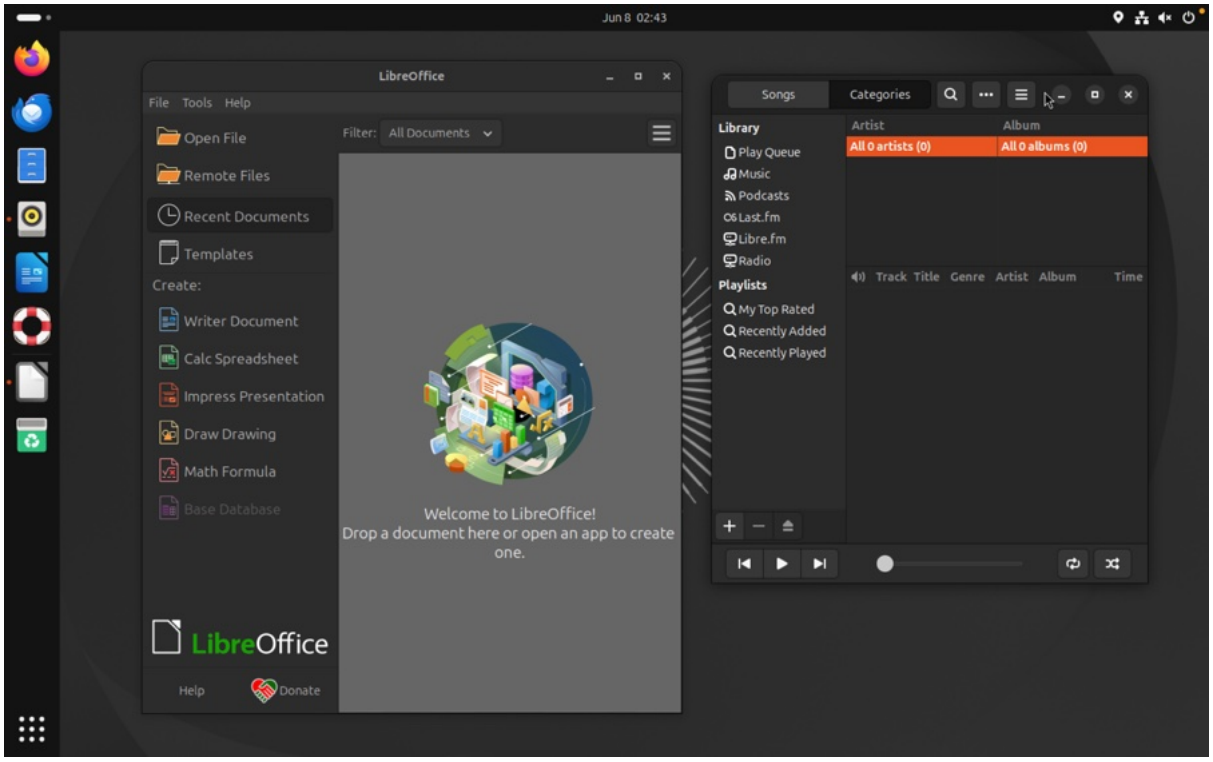
- Checking Ubuntu version help you install compatible software package.
- Checking Kernal version helps when troubleshooting broken hardware by showing exact driver and core operating system version.
- Checking Hostname helps you confirm which machine you are actually connected to(troubleshoot when managing multiple remote servers 'SSH').
- Checking Current User helps reveal current execution privileges troubleshooting sudo issues.
- Checking Current Directory helps pinpoint exact location in the file path/system(troubleshooting location issues).
- Checking Disk Usage helps highlight storage exhaustion troubleshooting data corruption, log and writes.
- Checking Network interface helps troubleshoot/diagnose local connectivity issues(down and up condition).

c) One Potential System default issue from my outputs is Internet connectivity although I have access to internet my network interface registered my connectivity as Ethernet enp0s1 when infect I'm connected to wireless internet(Wifi).

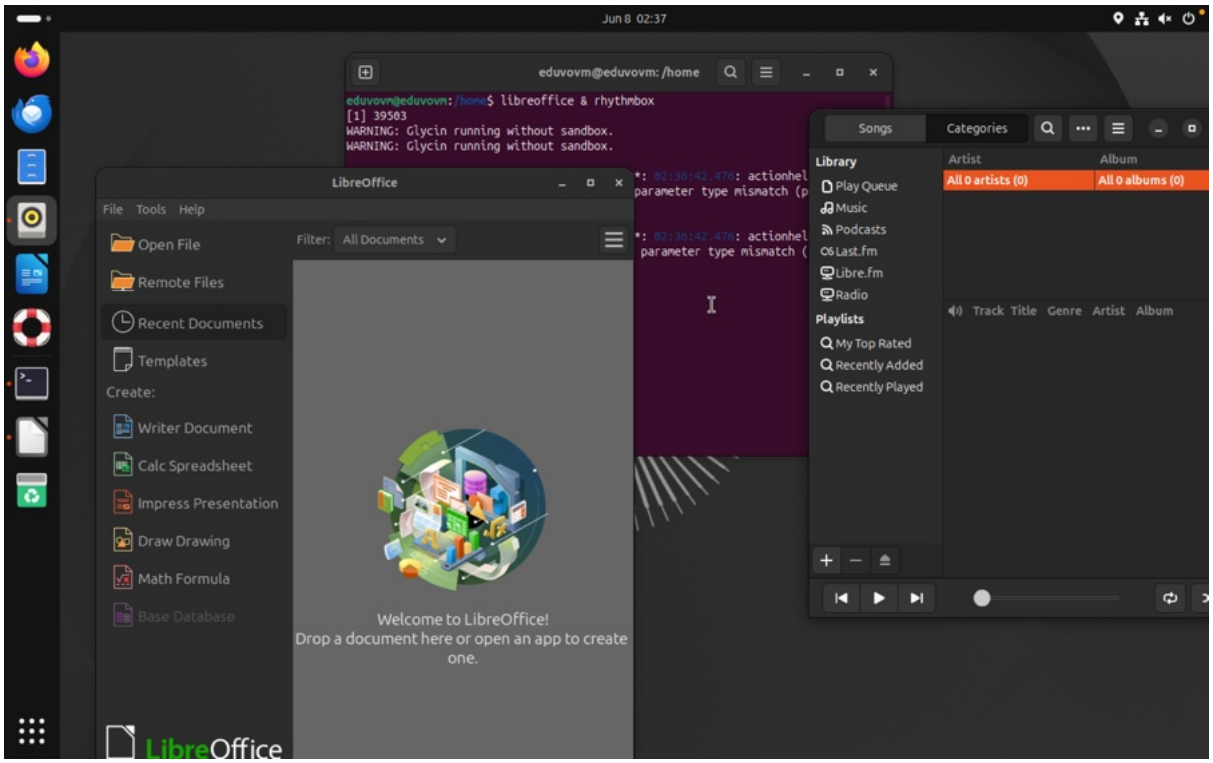
Question 3 GNOME Desktop and Preferences



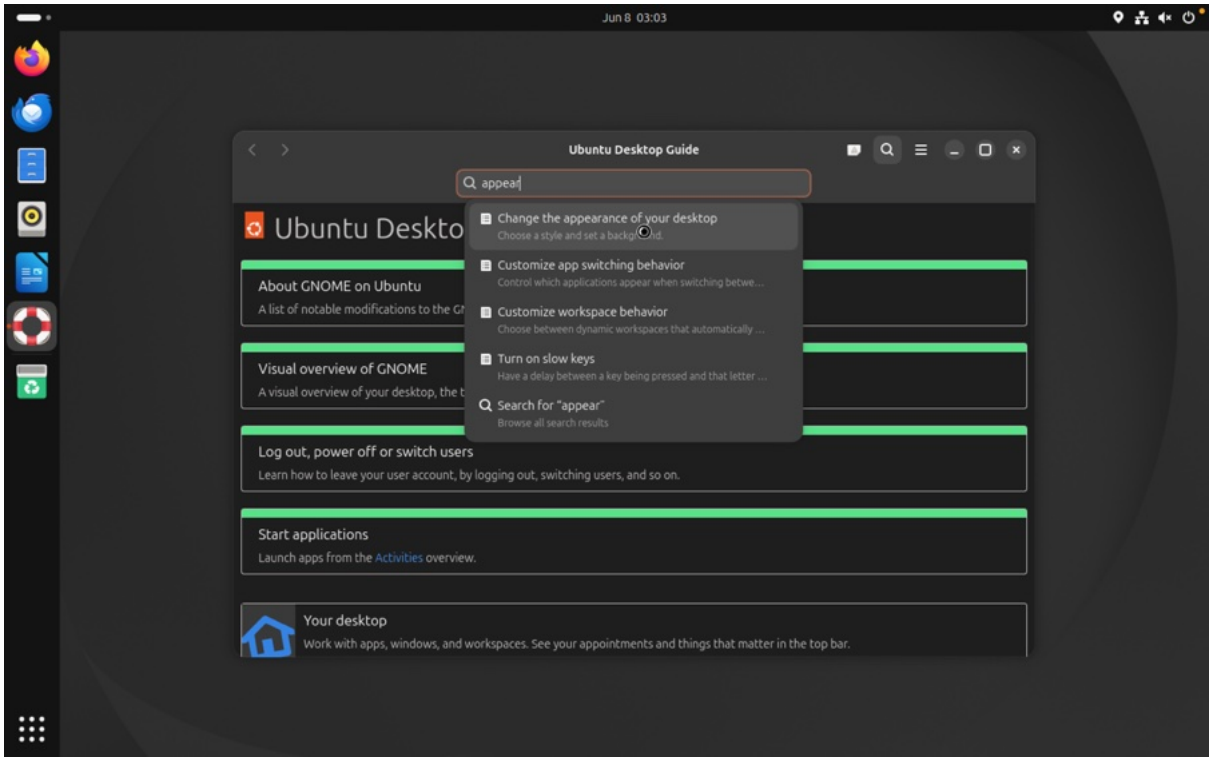


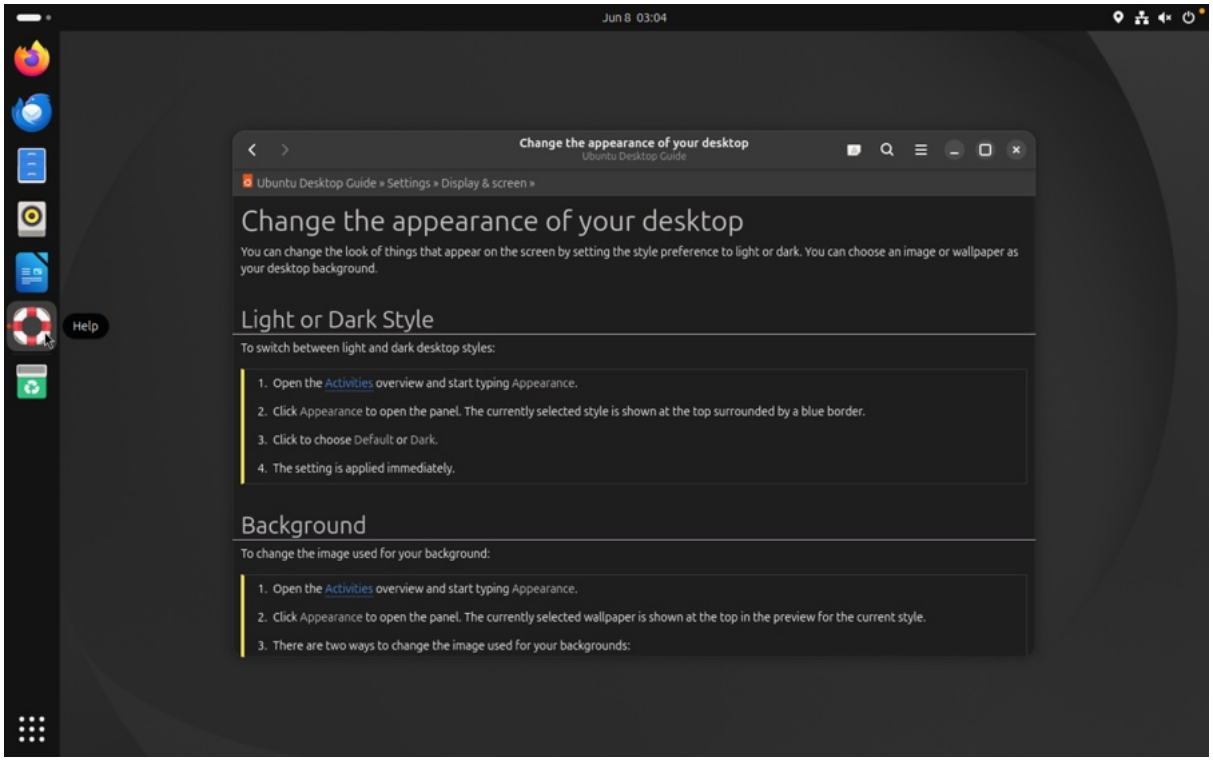


Launching TWO Application LibreOffice & Rhythmbox using the Ubuntu Graphical User Interface "select and click".

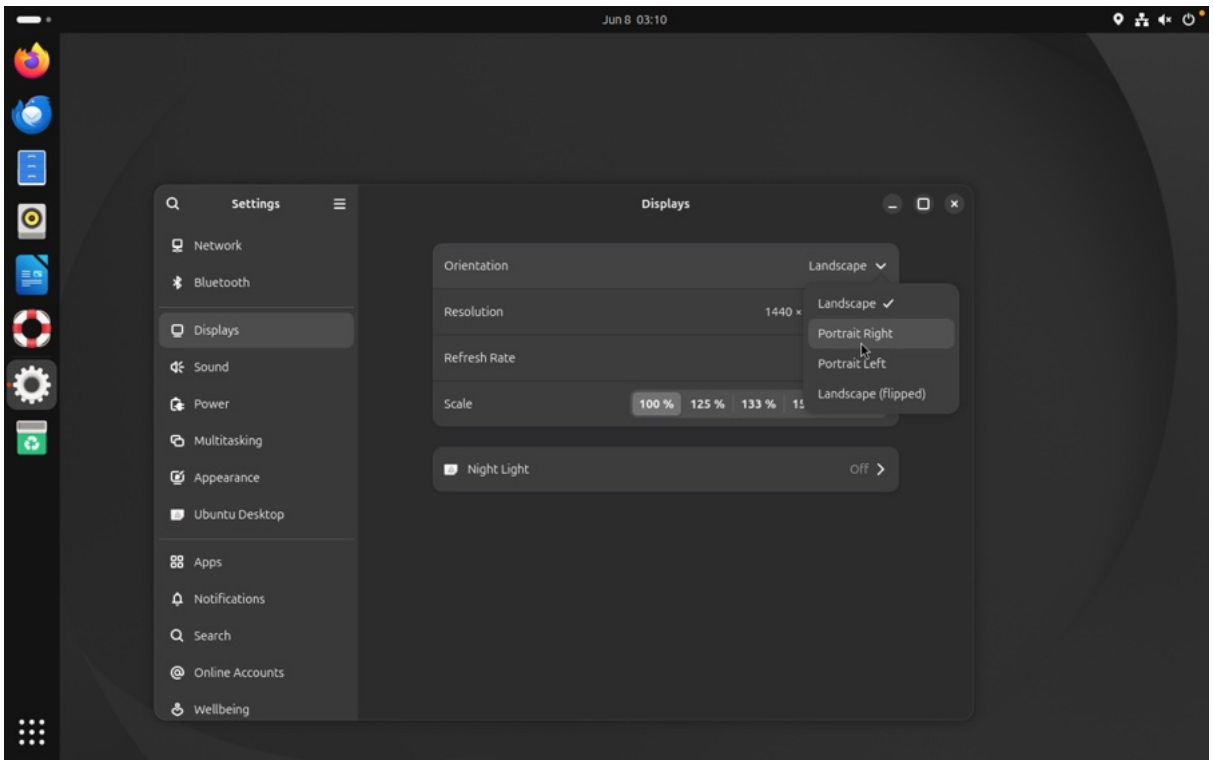


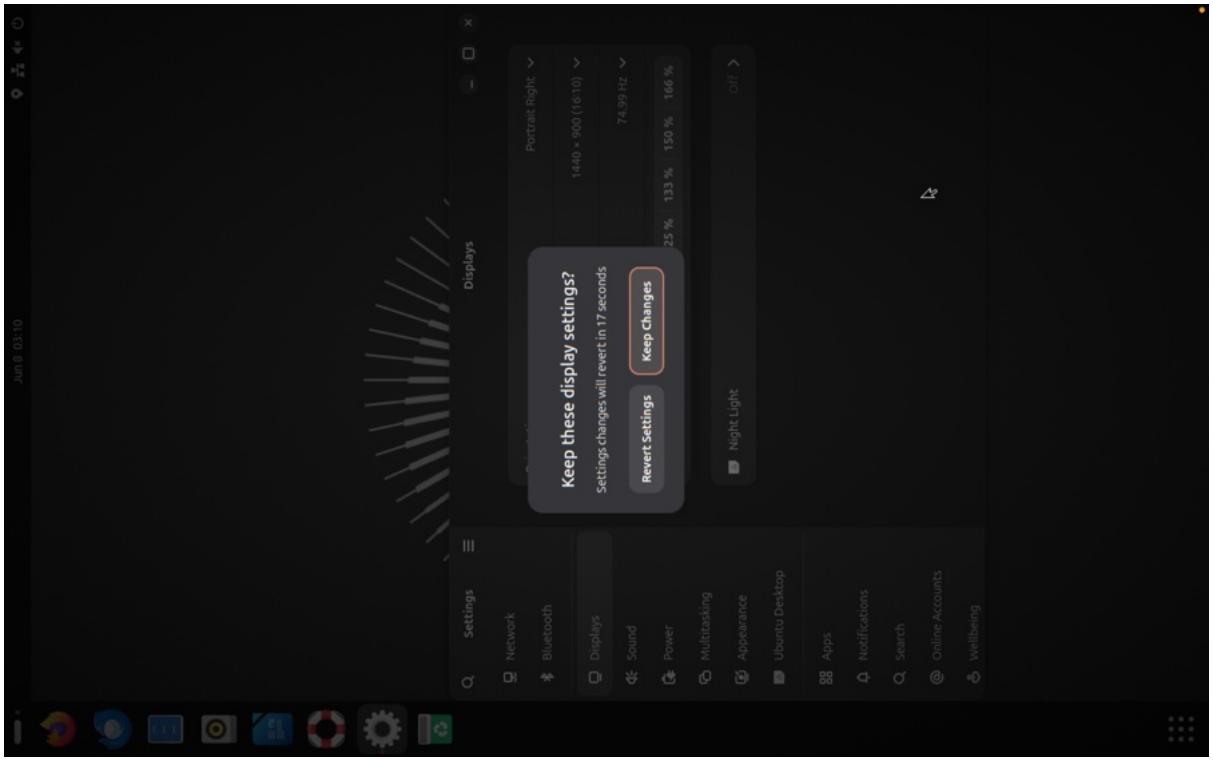
Launching TWO applications LibreOffice & Rhythmbox Using the Terminal and the & command.



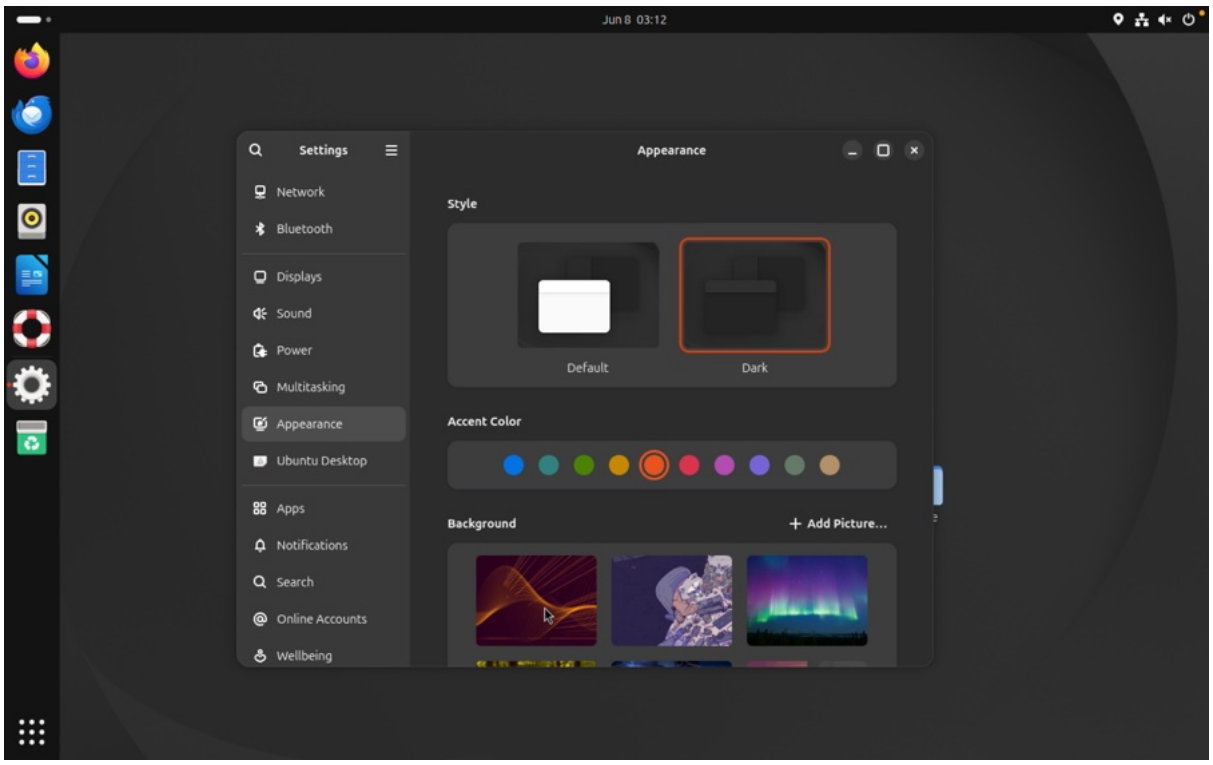


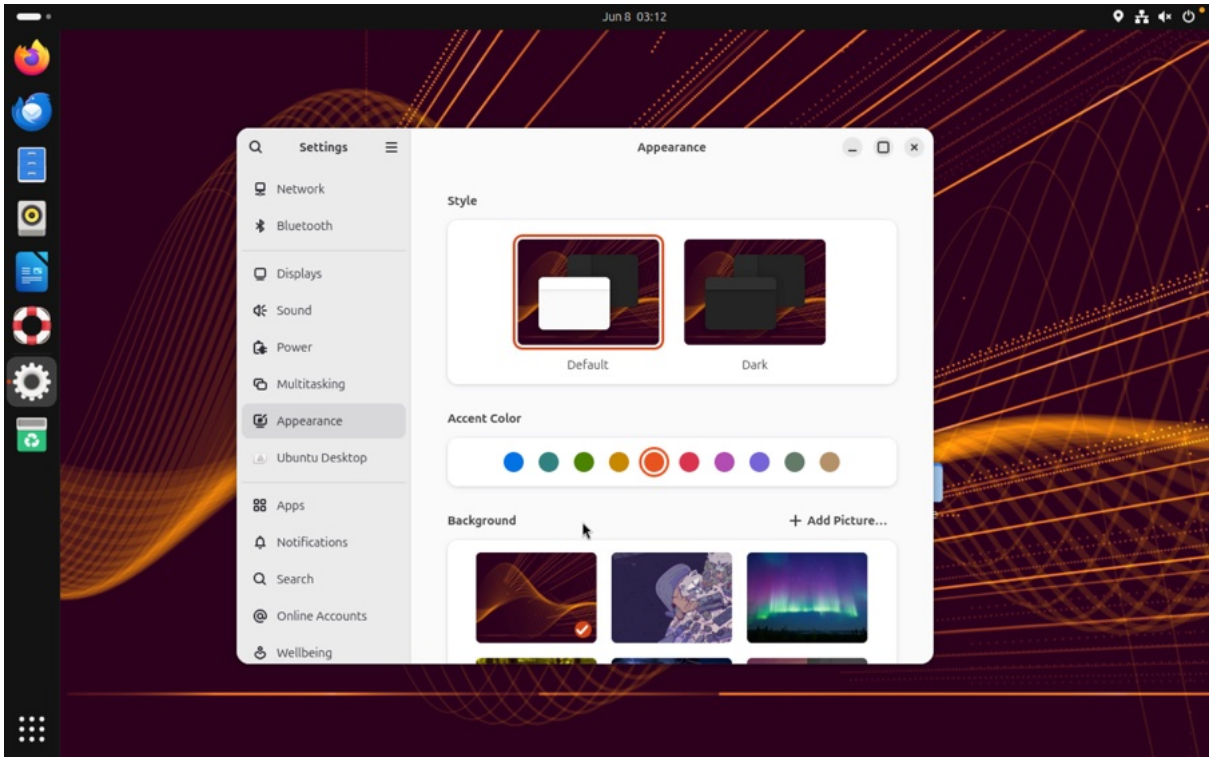
Using GNOME Help to search for the system feature Appearance





Modifying System Setting Display to change Landscape to Portrait Right.





Modifying System Setting Appearance to change Background and Style/Colour Mode(Dark/Light).

4) GNOME improves usability by prioritizing quick navigation in a Highly organized workspace driven workflow that's distraction free.

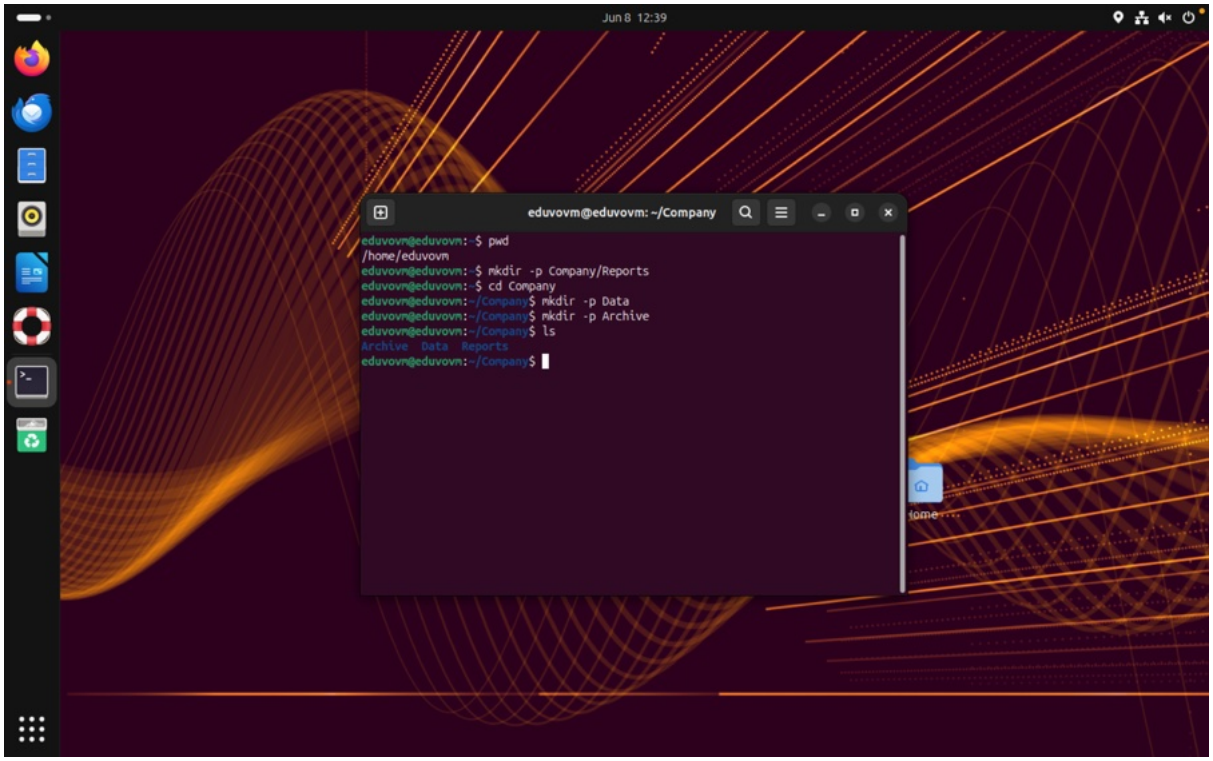
GNOME improved workflow usability and its strict design can frustrace a power using from like MacOS or Windows who may seek advanced customization.

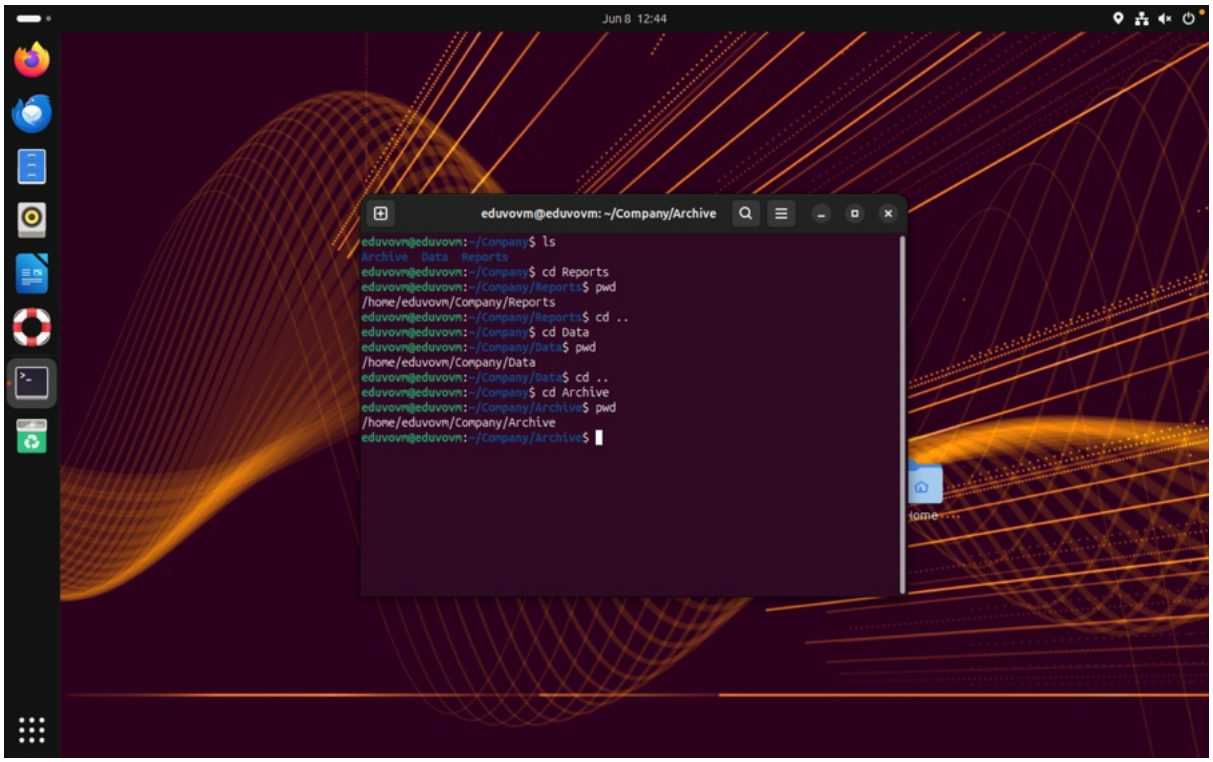
5) My System changes affects system usability by changing orientation from Landscape to Portrait Right which affects my display in a Negative way as my laptop software and hardware is designed for Landscape usability.

And my other change is Appearance background and style which make the OS more bright and engaging.

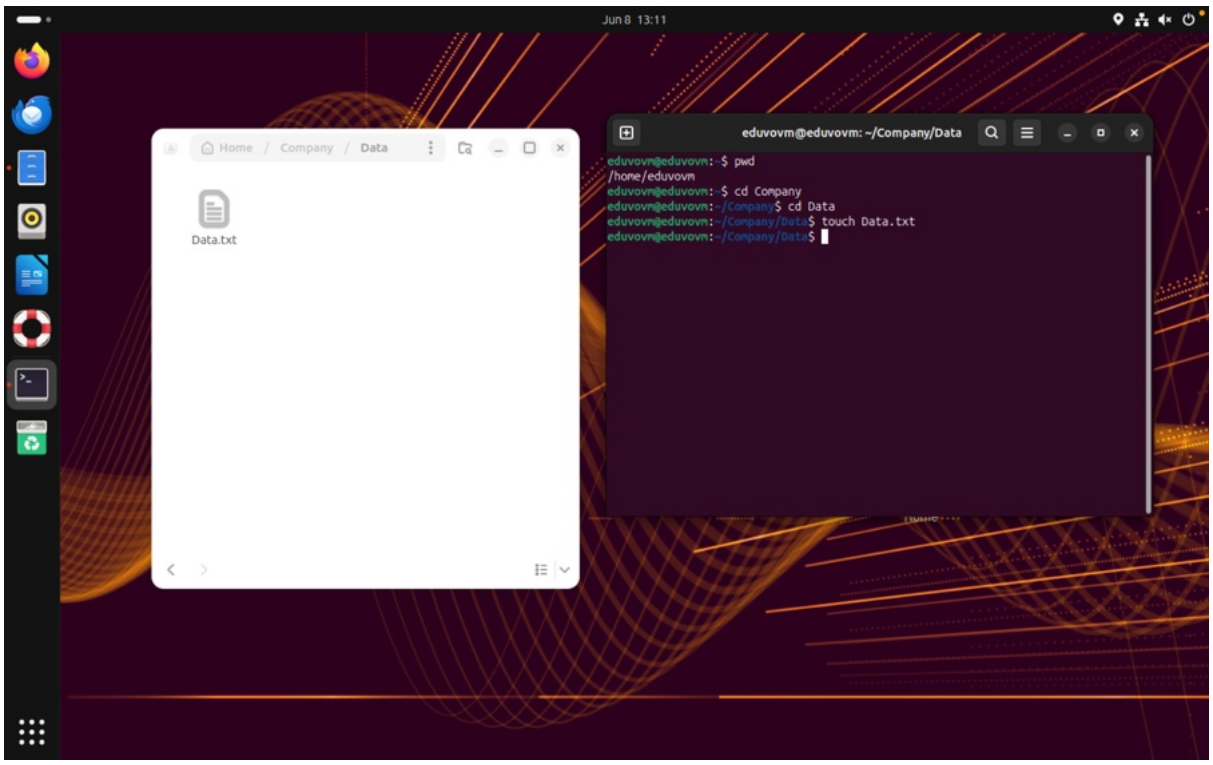
6) Graphical User Interface is better than Command Line Interface which looking for. Visual context, ease of use or unfamiliar with software.

Question 4 File System and Directory Management

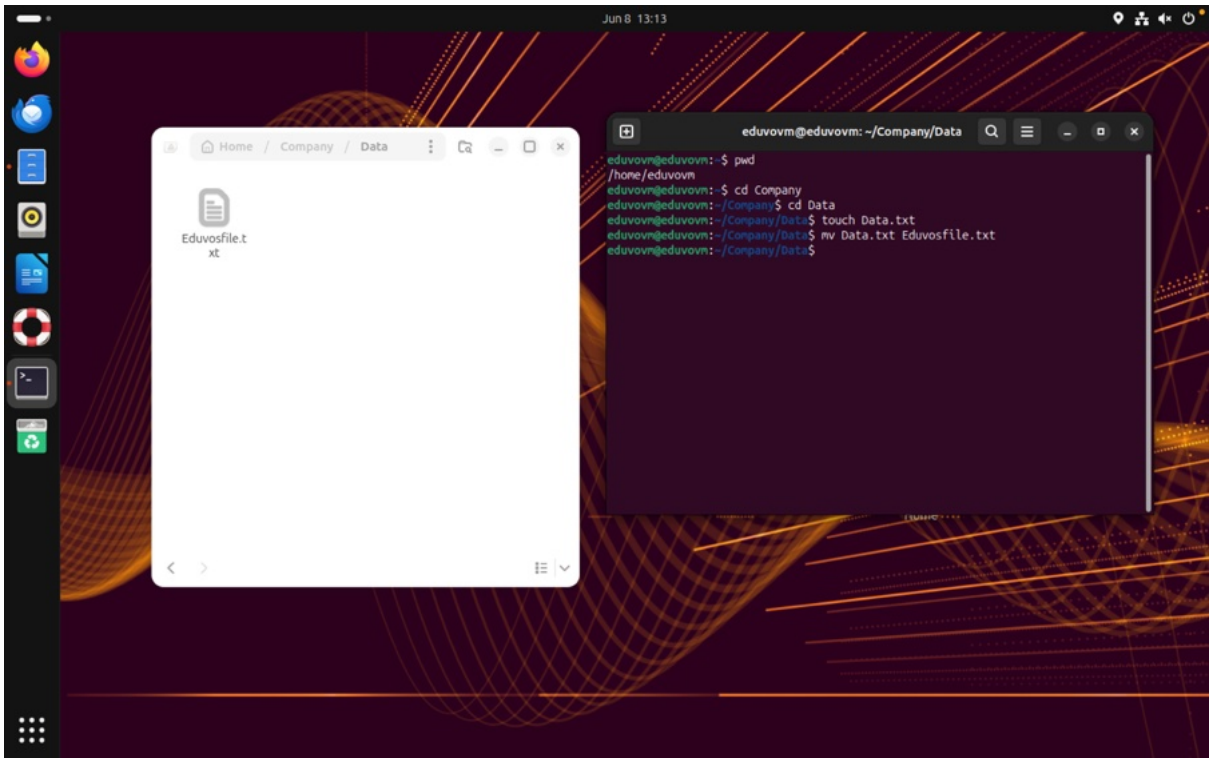




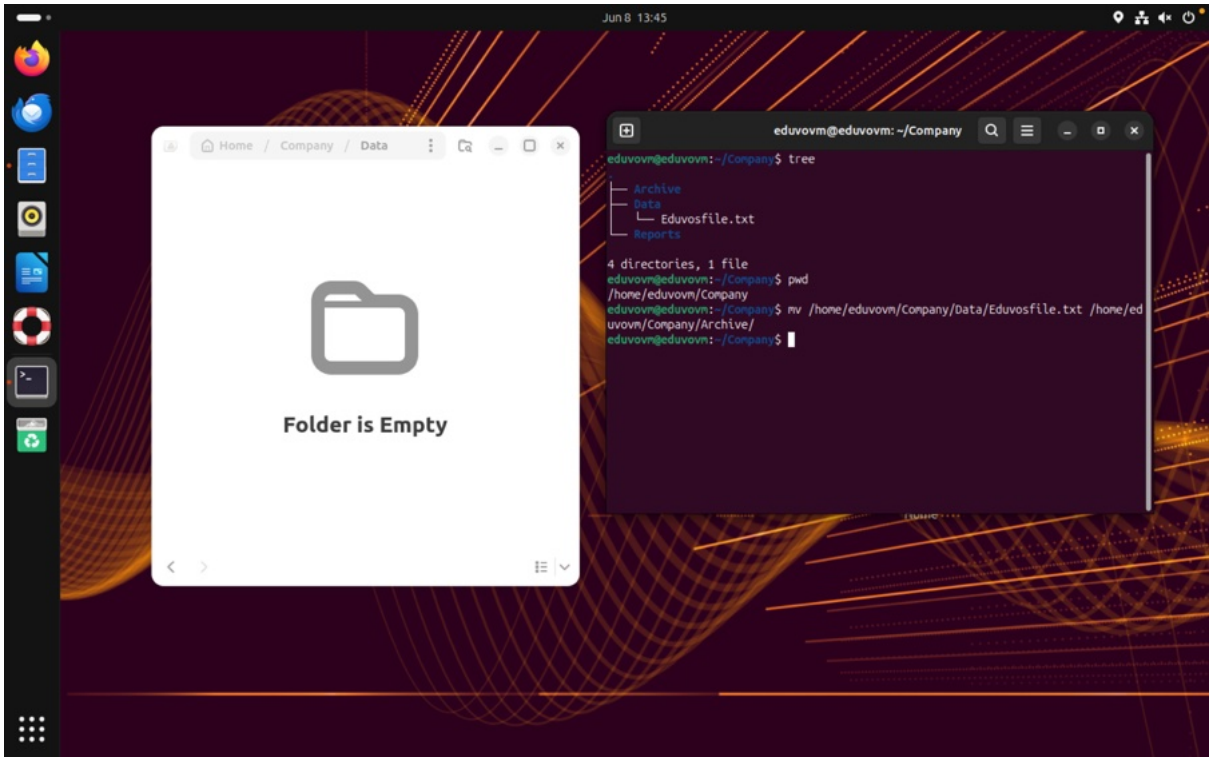
Creating Folder/Directory Structures/: Company/(Reports, Data and Archive)

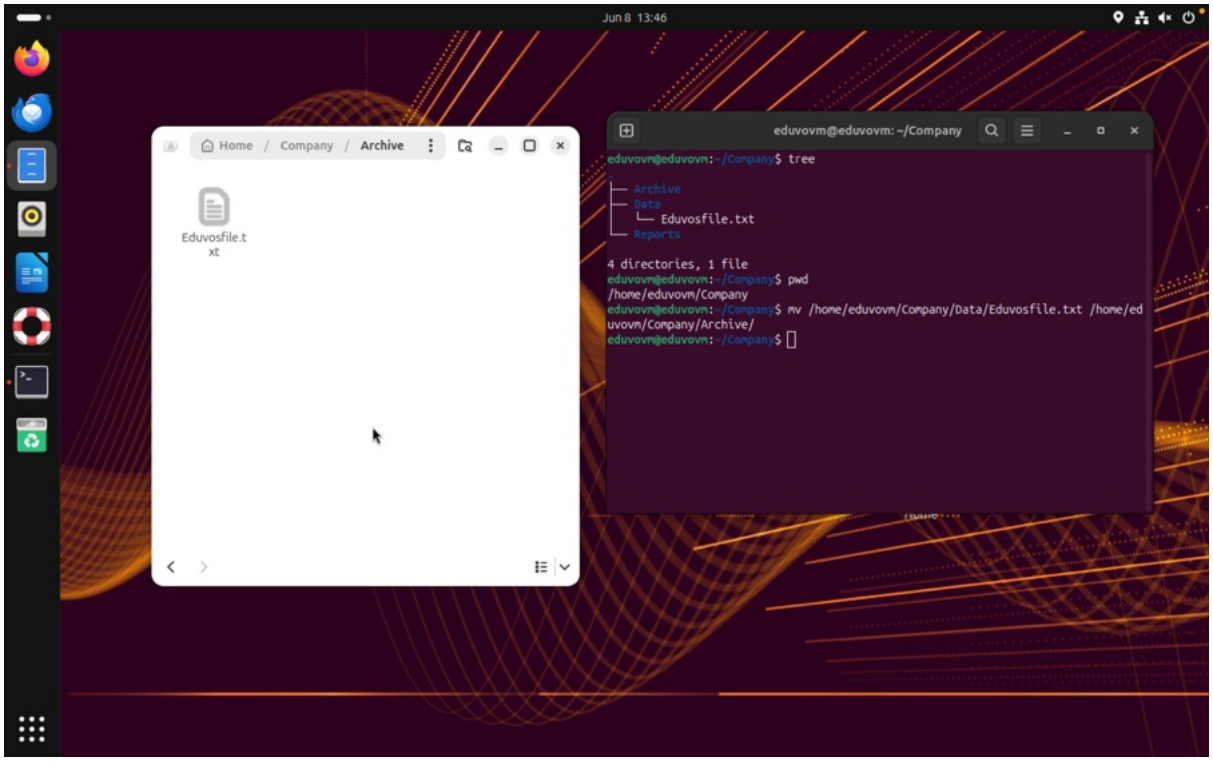


Creating File Data.txt using Command touch.

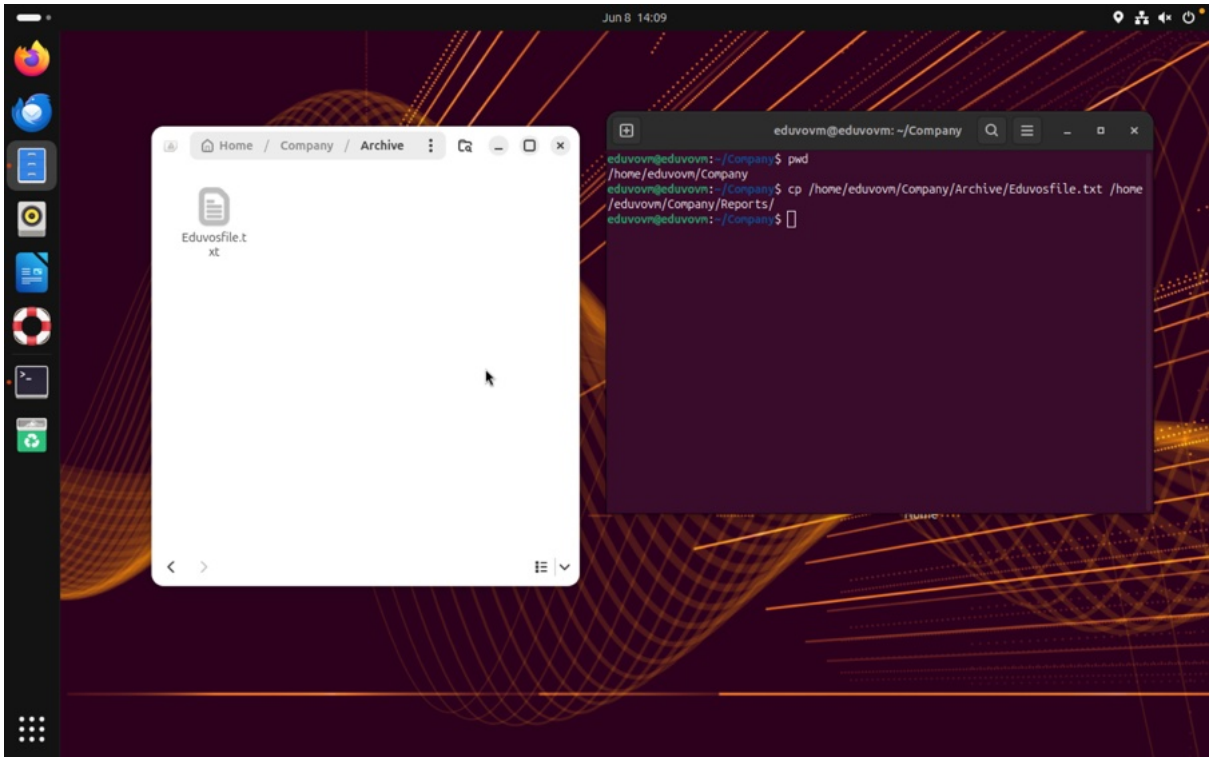


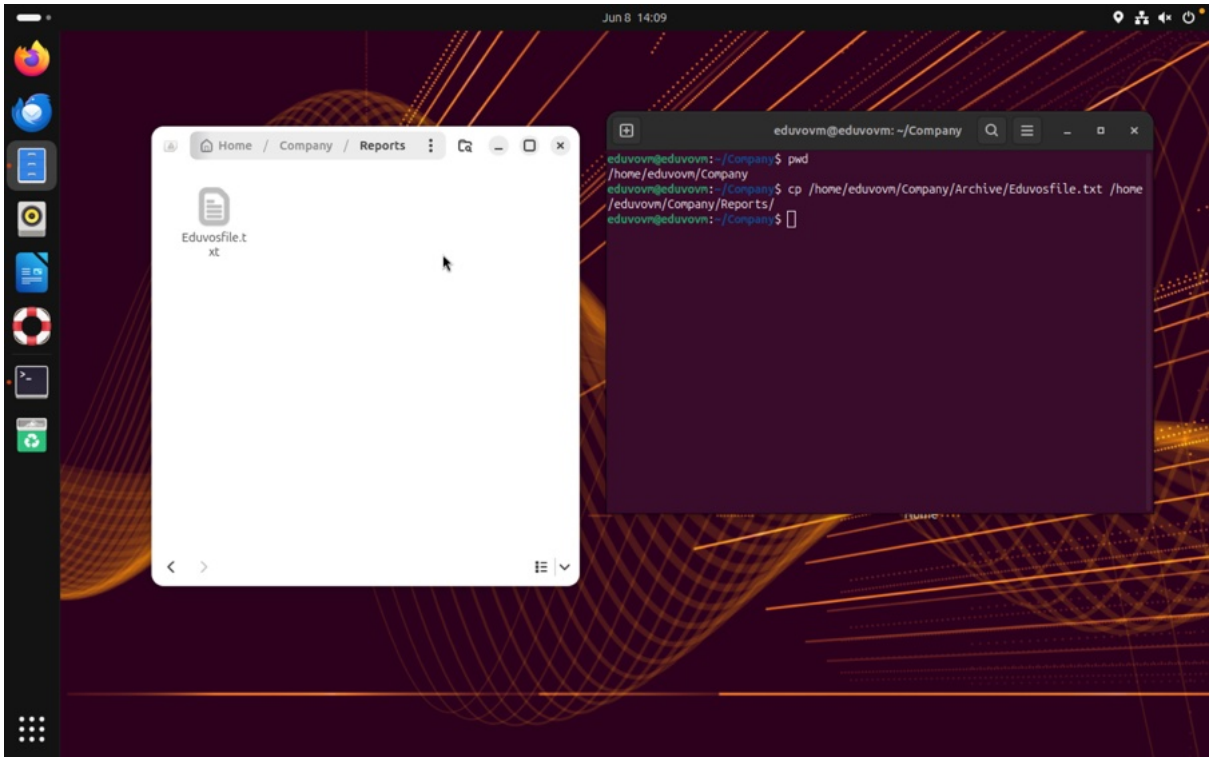
Renaming File Data.txt to Eduvosfile.txt using Command mv.



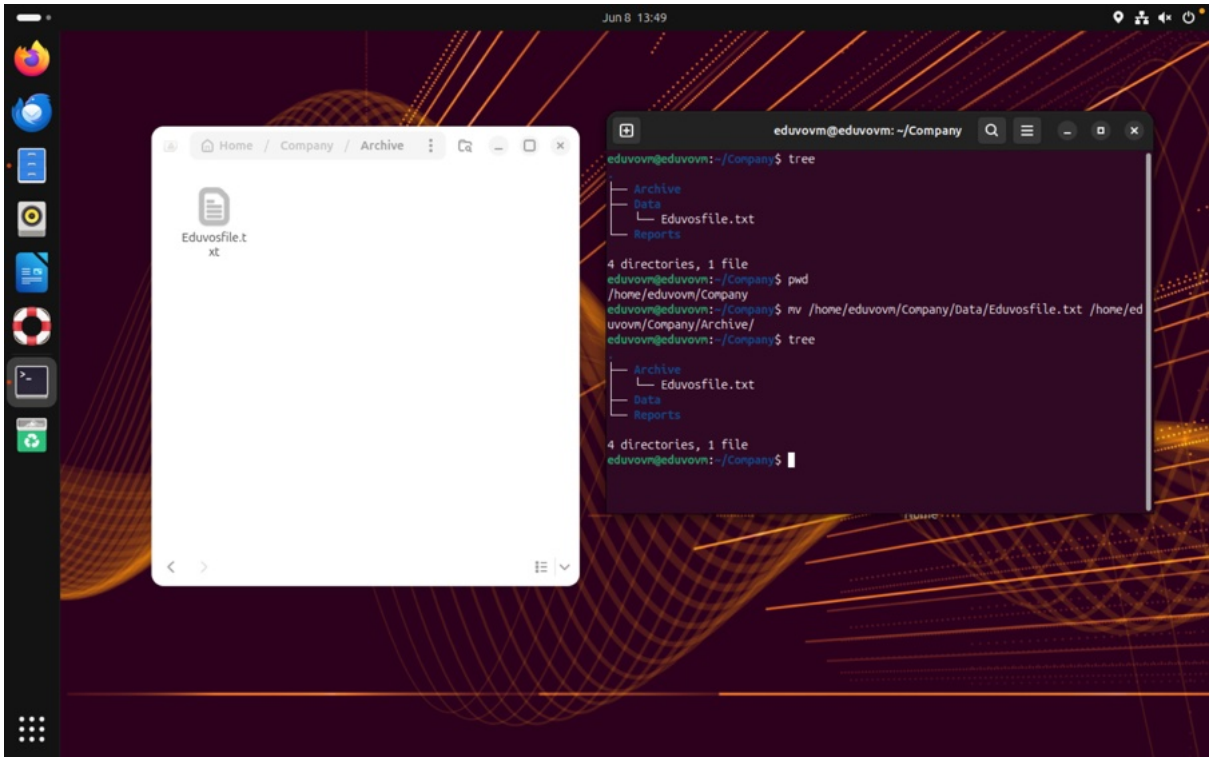


Moving File Eduvosfile.txt From Data to Archive folder using Command mv.

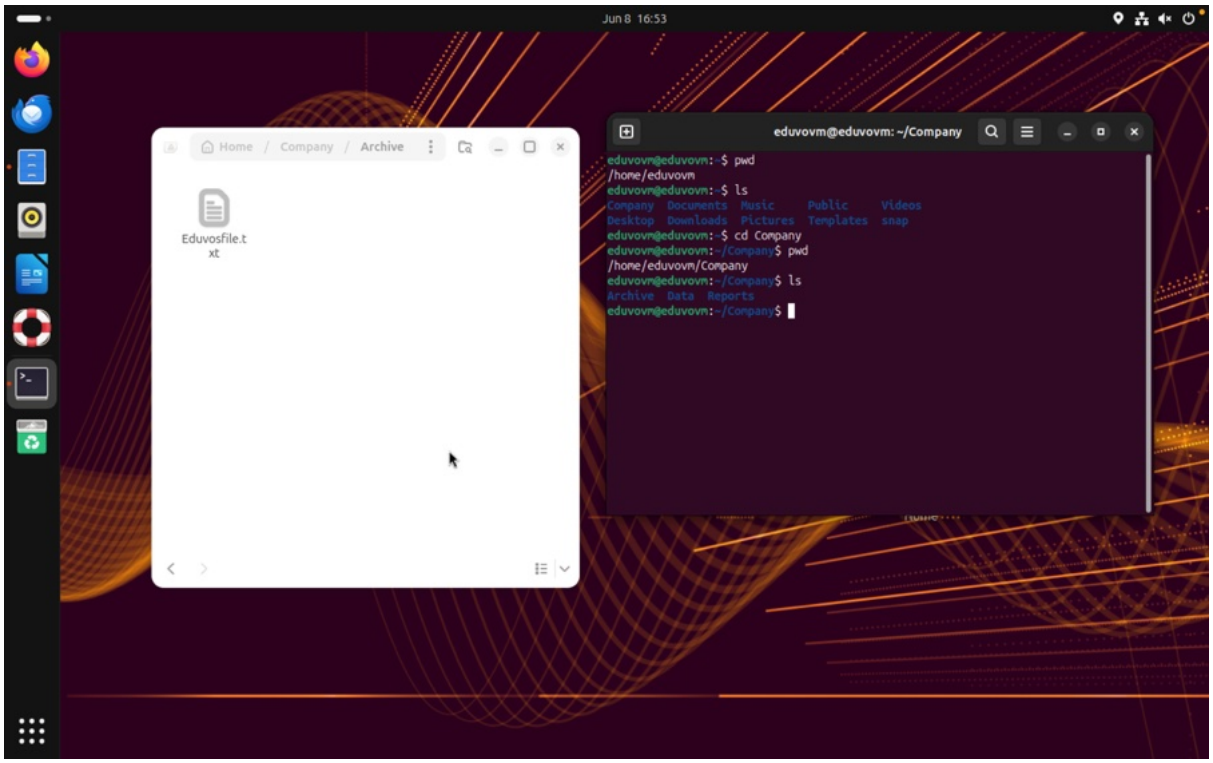




Copying file Eduvosfile.txt in Archive folder to Reports folder using Command cp.



In this structure of moving my file I've made a great example of Absolute Path as it specifies the complete location from the absolute start of root, using "/"syntax. Eg "/home/eduvovm/Company/Data/Eduvosfile.txt".



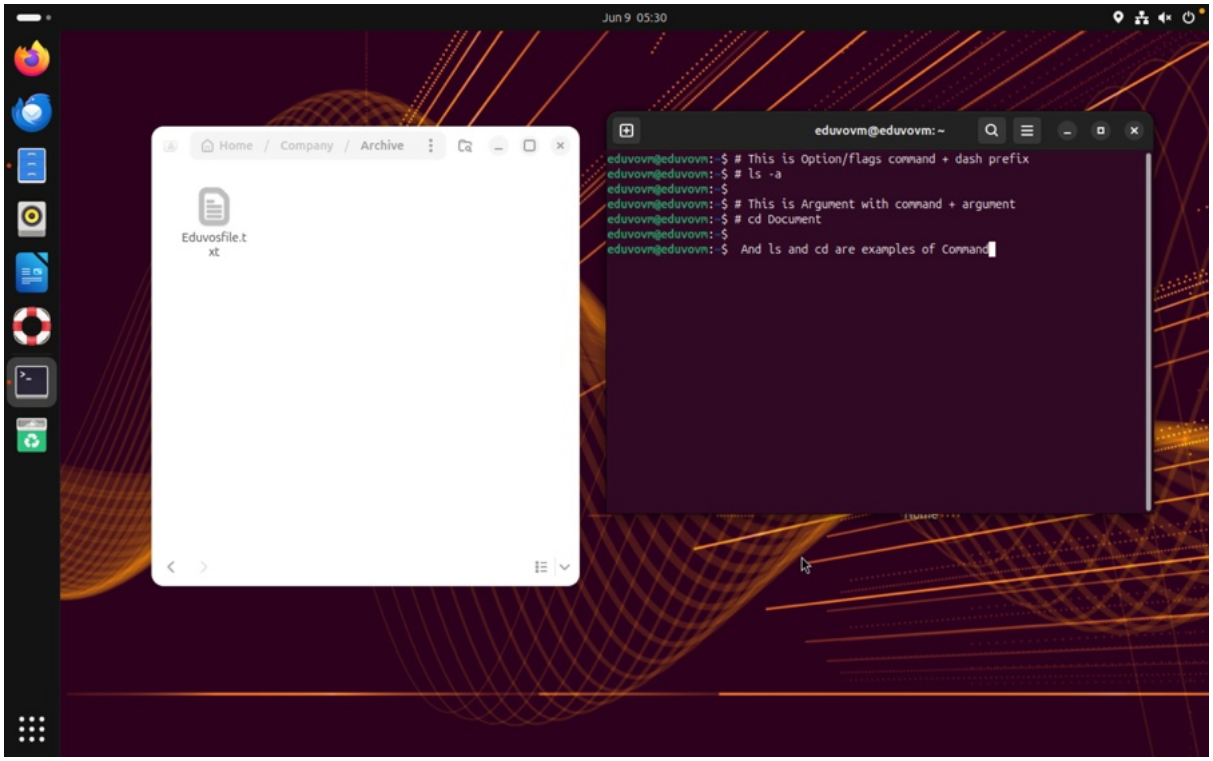
This Structure navigating in the terminal is a great example of Relative Path as it shows that relative path is a path that depends on your current location(shown by pwd) which navigate terminal quickly and may use single or double dot to path current and parent directories.

3) Linux uses a single-root filesystem to provide storage abstraction which allows all files, directories and physical devices to be organized in a single predictable hierarchy.

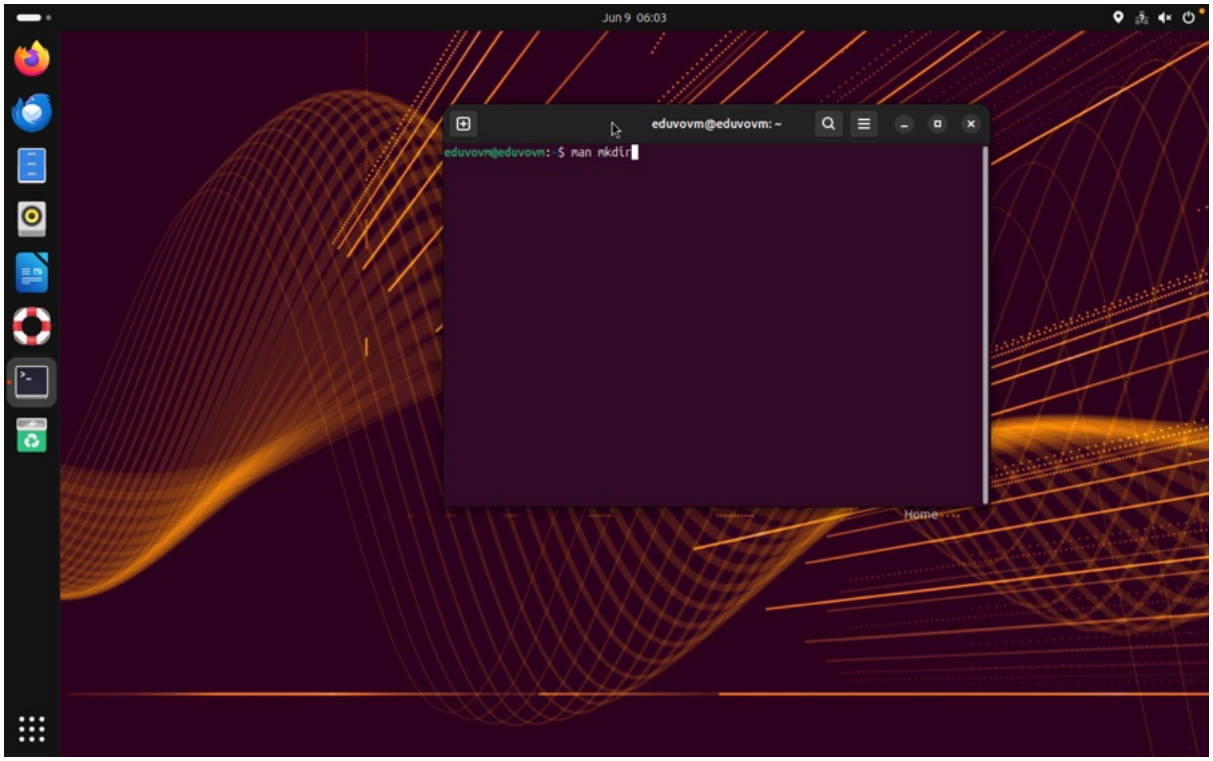
In which so programs can rely on absolute file paths without worring about physical disk partition.

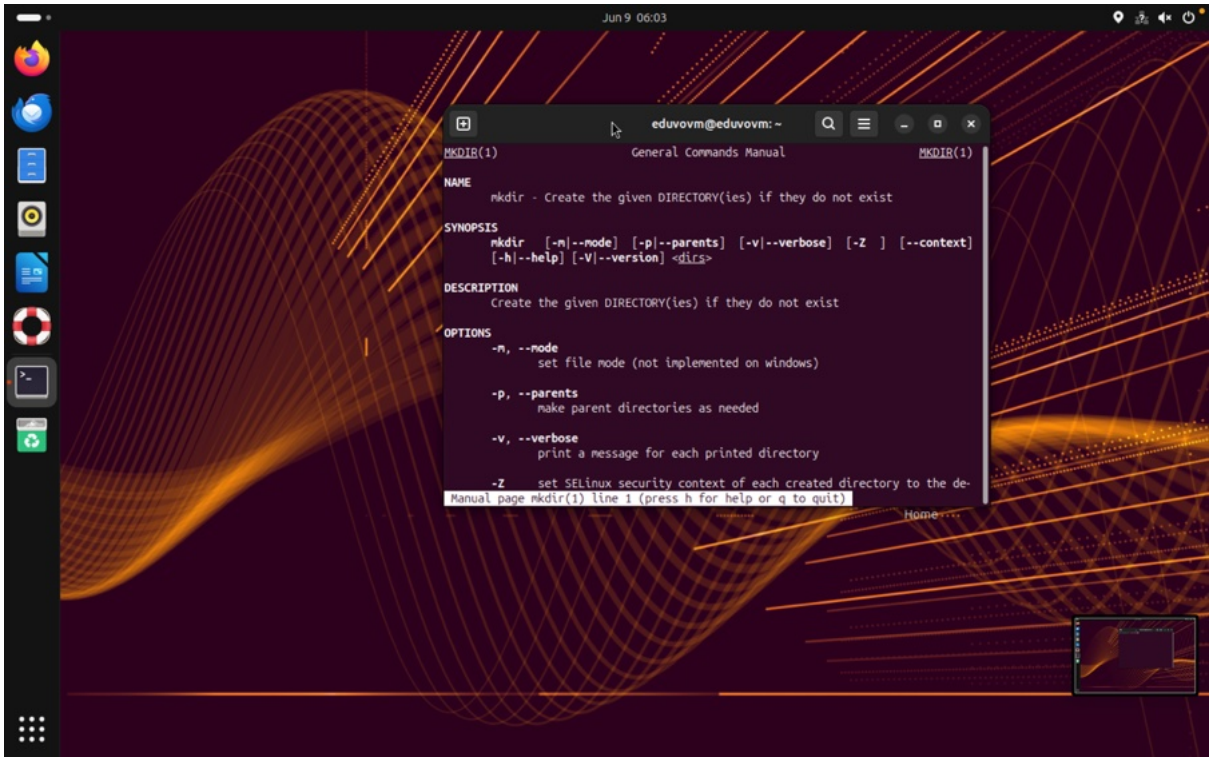
4) Risks of incorrect file handling includes Data Loss and Security Vulnerabilities. Where as with data loss command like rm -rf can wipe entire system root directory. And with security vulnerabilities executing broad permission on sensitive systems directories or private keys like can expose sensitive data to malware or unauthorized local users.

Question 5 Terminal and vi Editor

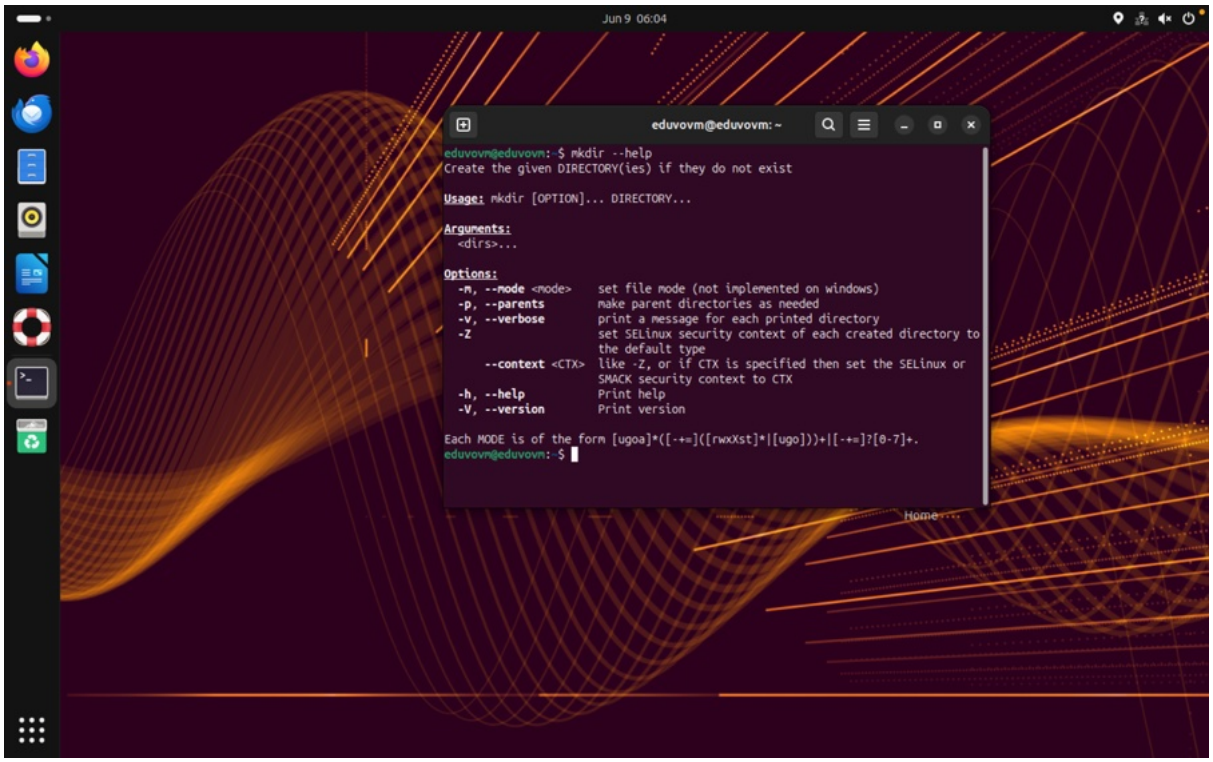


The command syntax of Ubuntu Linux is shown with following properties of Command, Option/flags and Arguments. With ls and cd being the commands, -a being the Option and Document as the Argument.

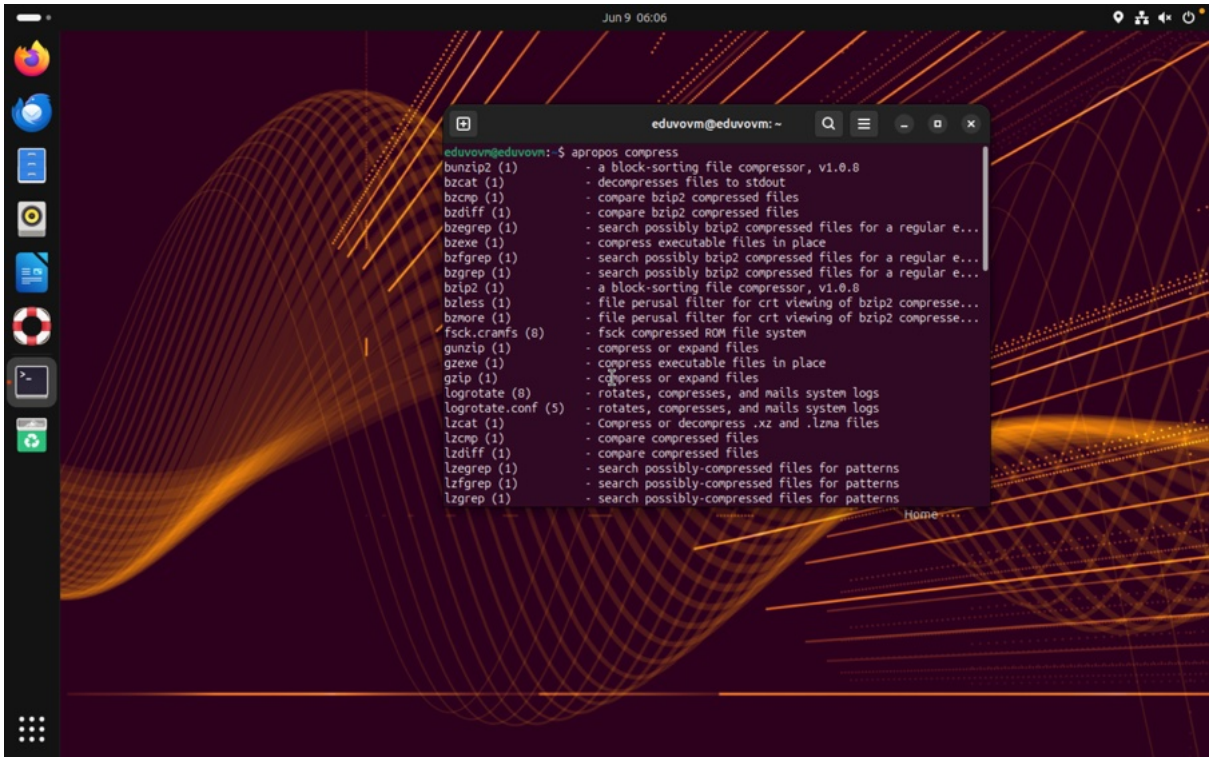




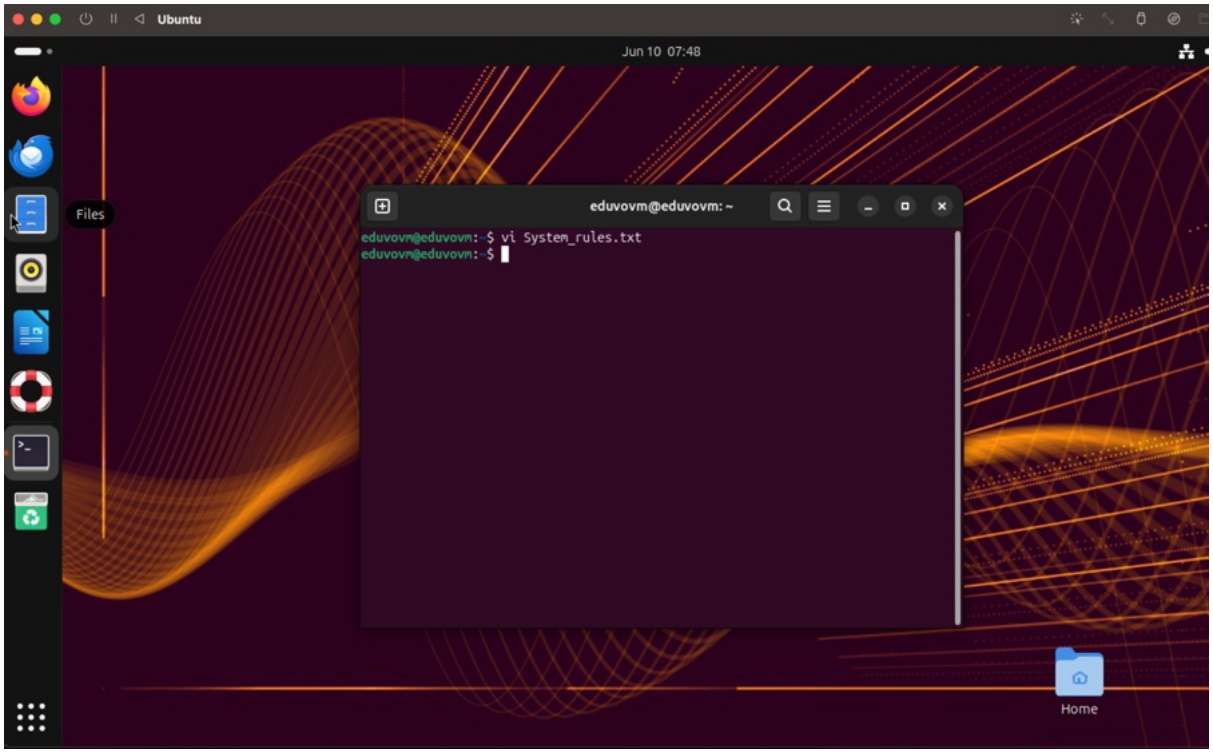
Using man to open the official manual for the mkdir command.

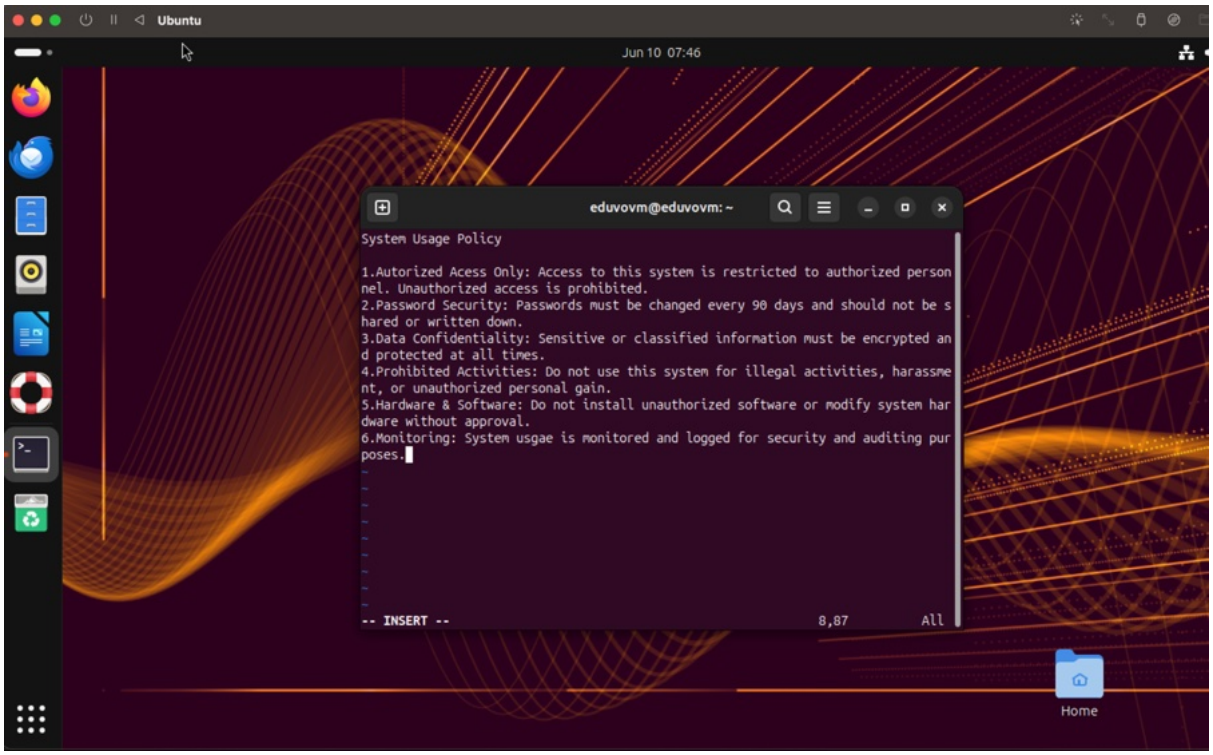


Using --help to request a usage summary of the command mkdir .

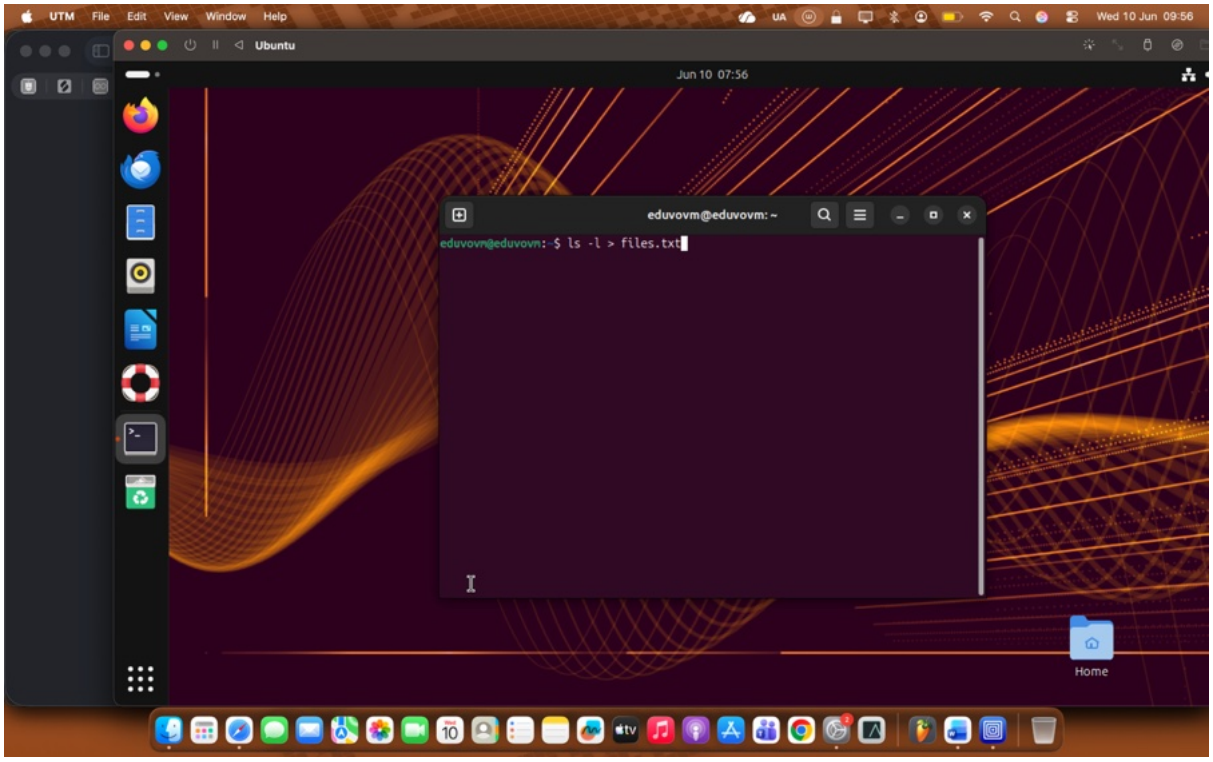


Using apropos to search local database for commands that can help me with compress properties.

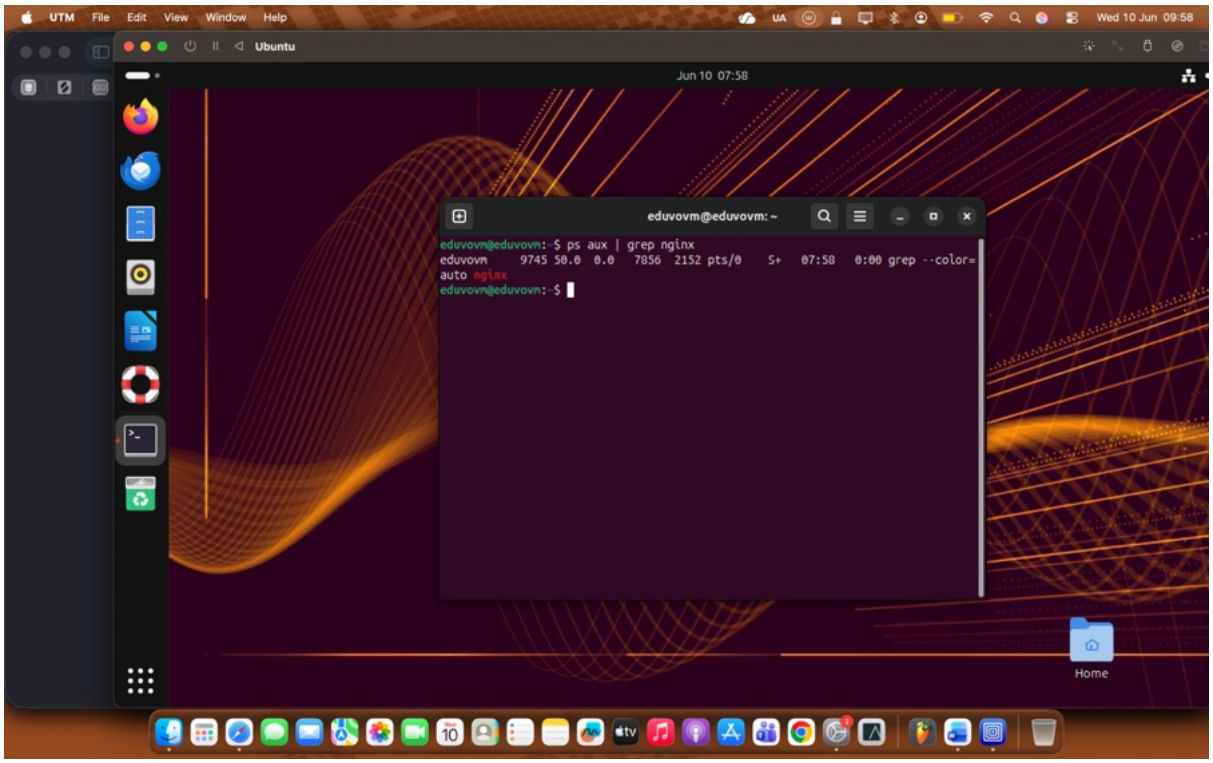




Creating and editing a file containing System File Rules with the title System Usage Policy using vi .



Using Output redirection to alter the text to go to file.txt text file with >.



Using Pipe to connect commands ps and grep together with x | x.

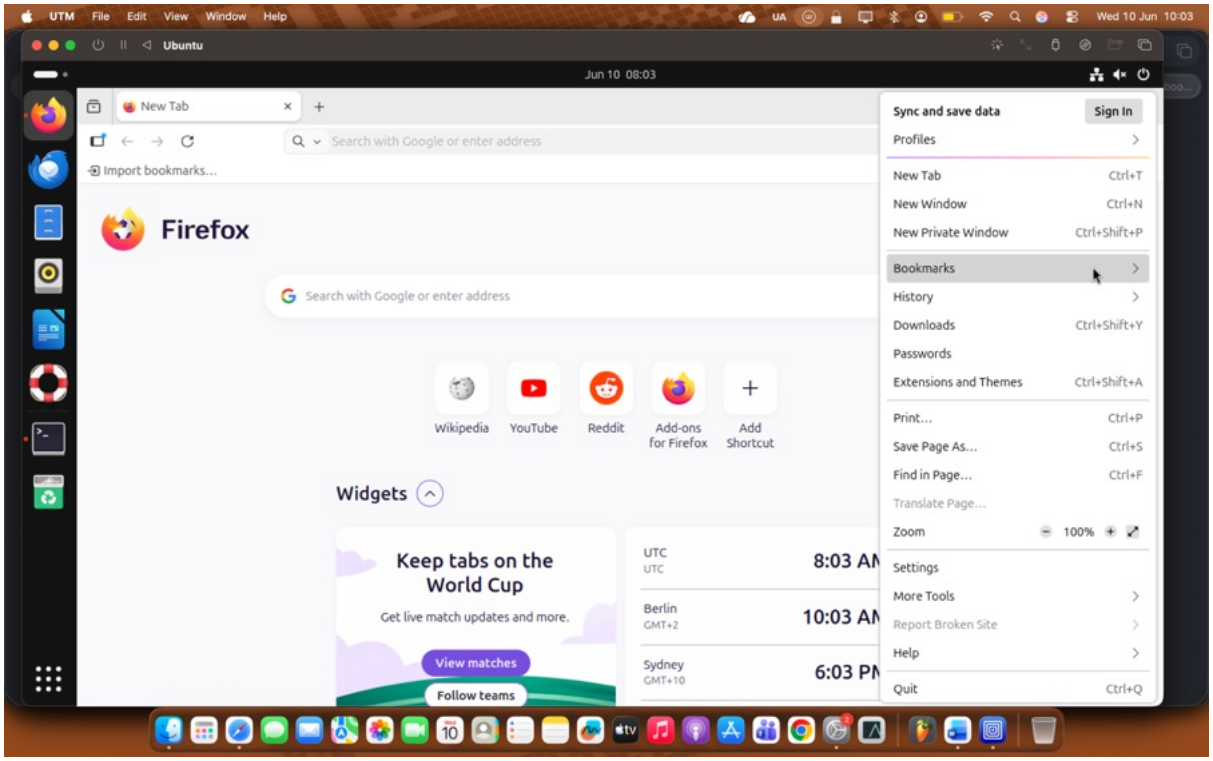
5) man, --help and apropos are all similar in the sense that they all help users to use commands efficiently in different way as man provides a complete manual for a specific command, --help provides a quick summary for a specific command and apropos searches commands you can use to achieve a specific goal.

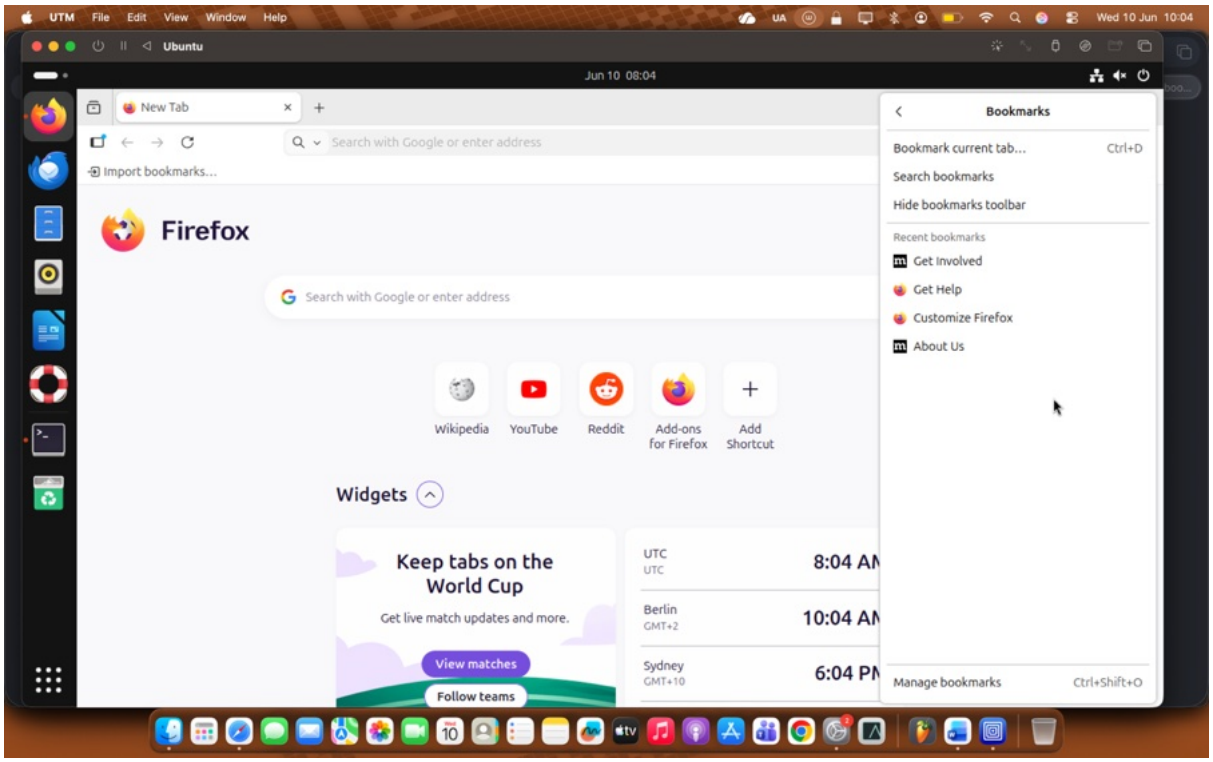
6) Vi modes are distinct state of operation in the vi text editor.

In which Vi separates navigation from text entry to maximize keyboard efficiency.

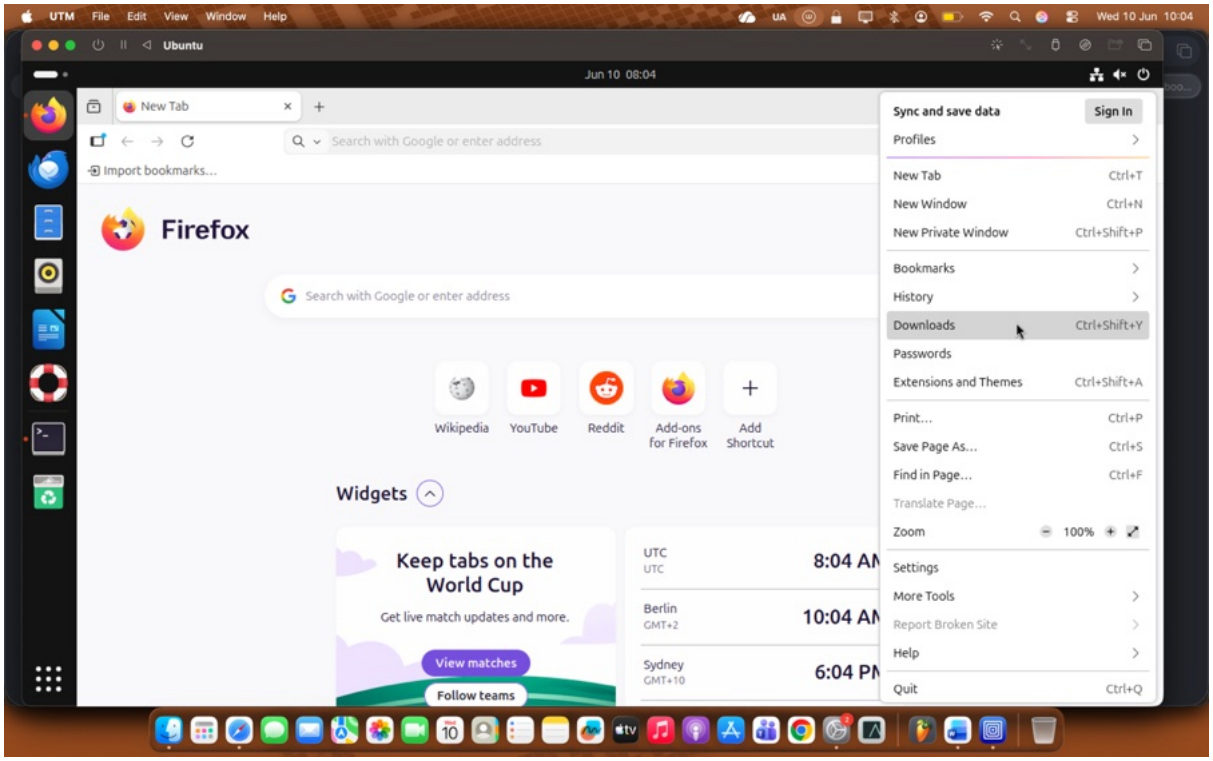
7) The difference between Pipe and Redirection is that Pipe connect the output of one program to the input of another, allowing programs to chain together. And Redirection connects a programs input or output to a file or stream, enabling save or load data.

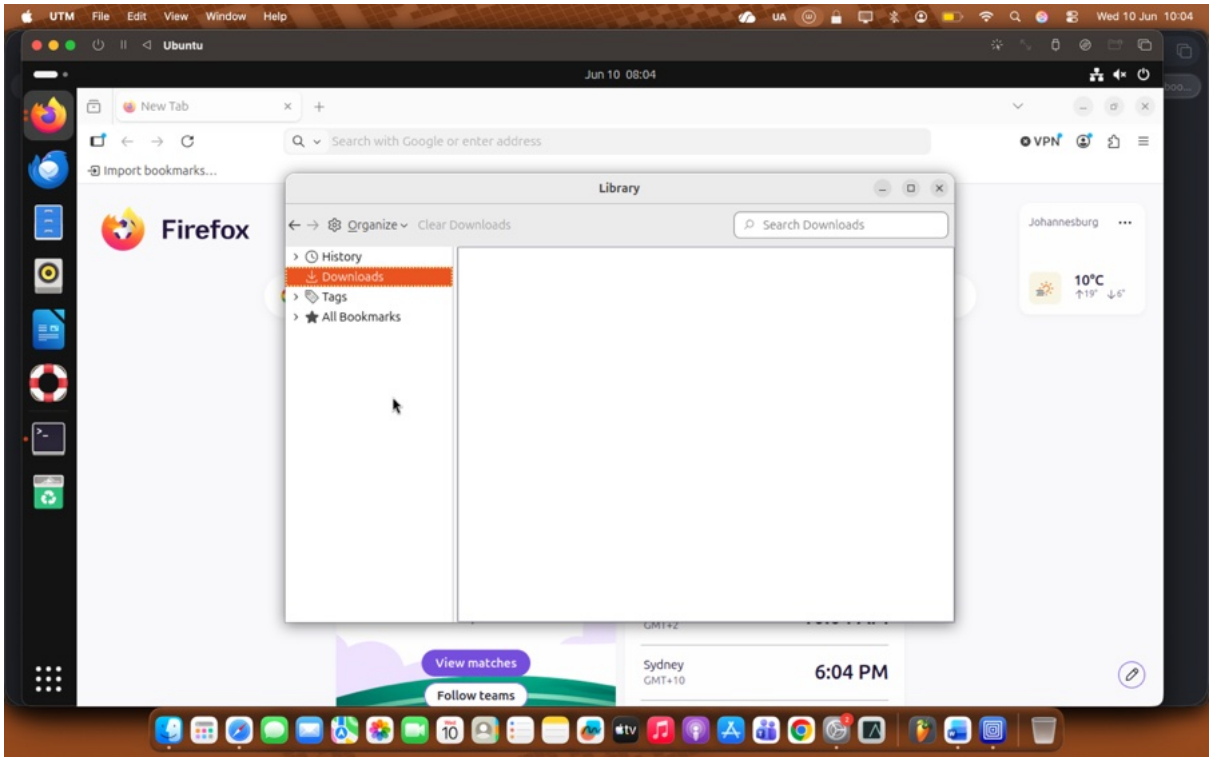
Question 6 Terminal and vi Editor



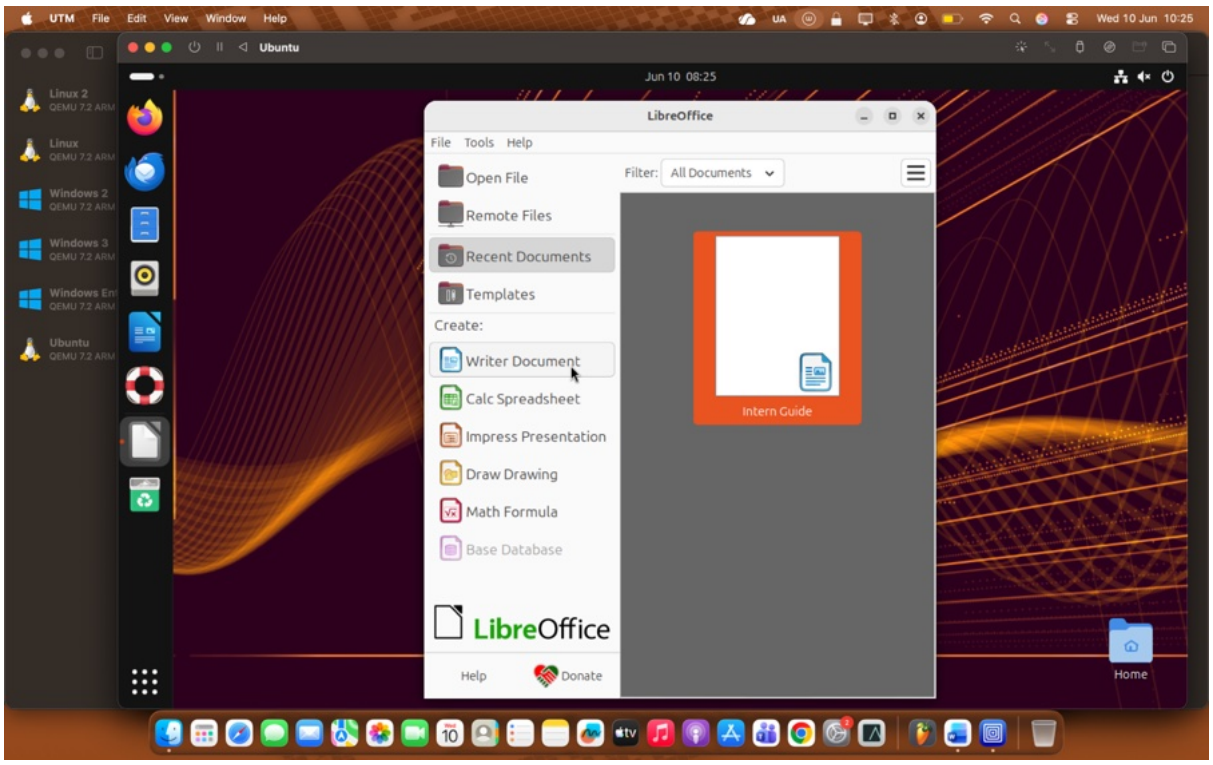


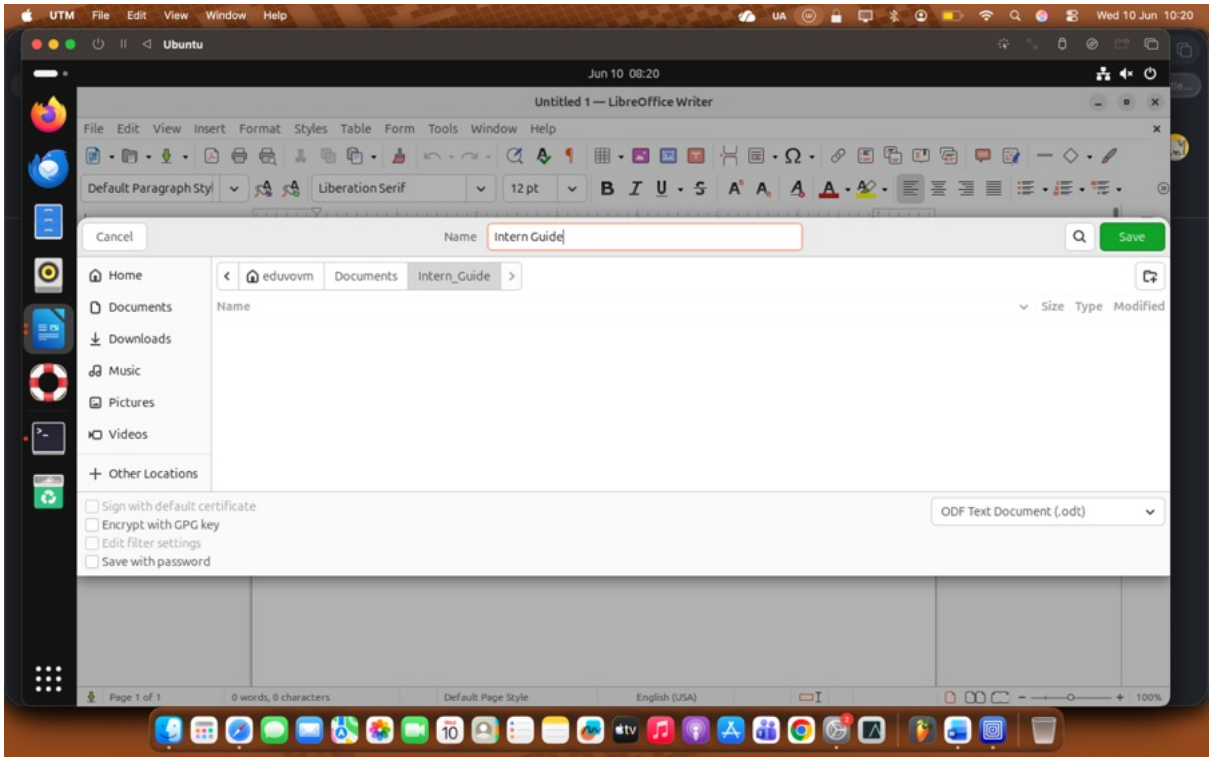
Using FireFox to locate and open bookmark

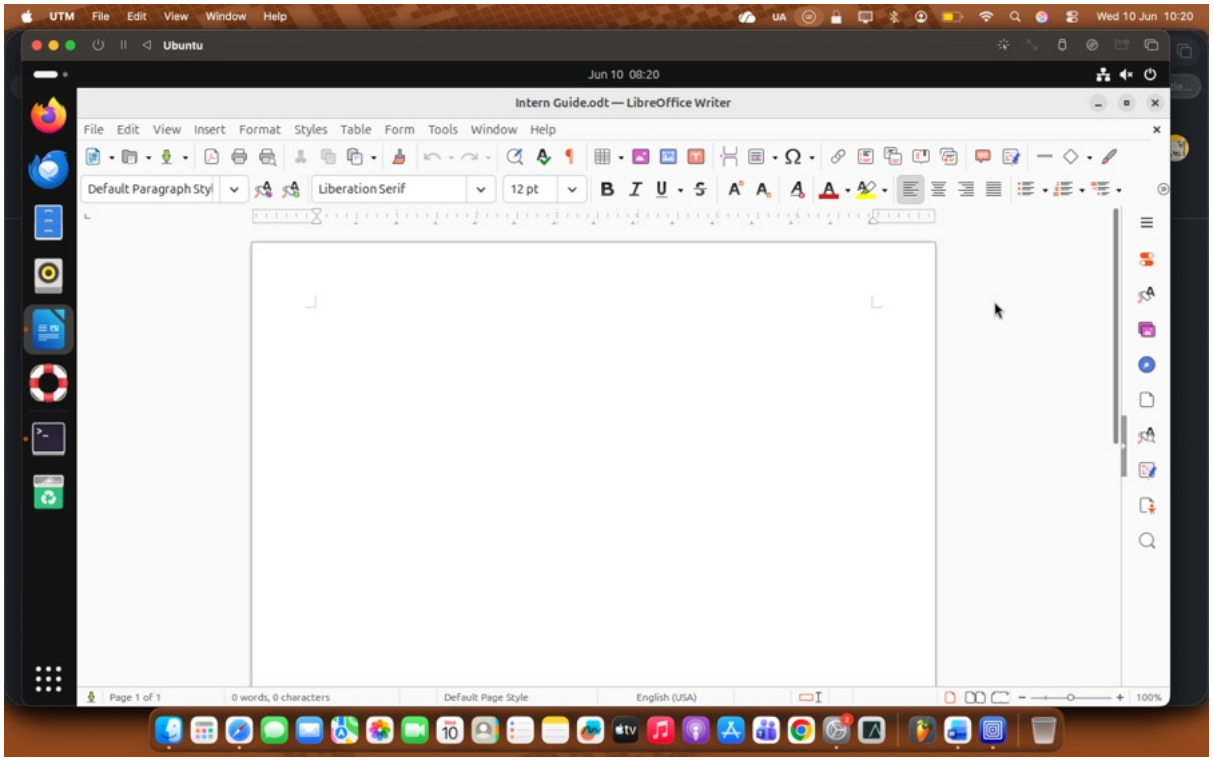




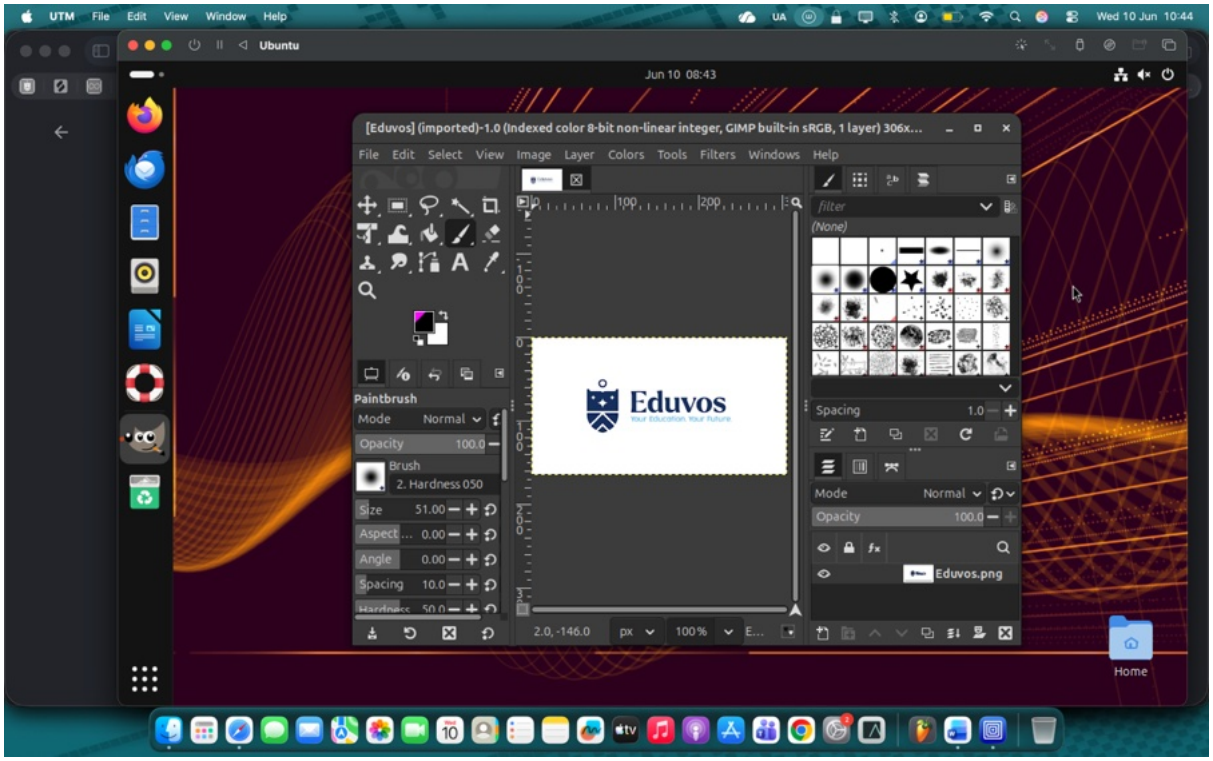
Using FireFox to locate and open Download and access library to organise downloads and bookmarks.

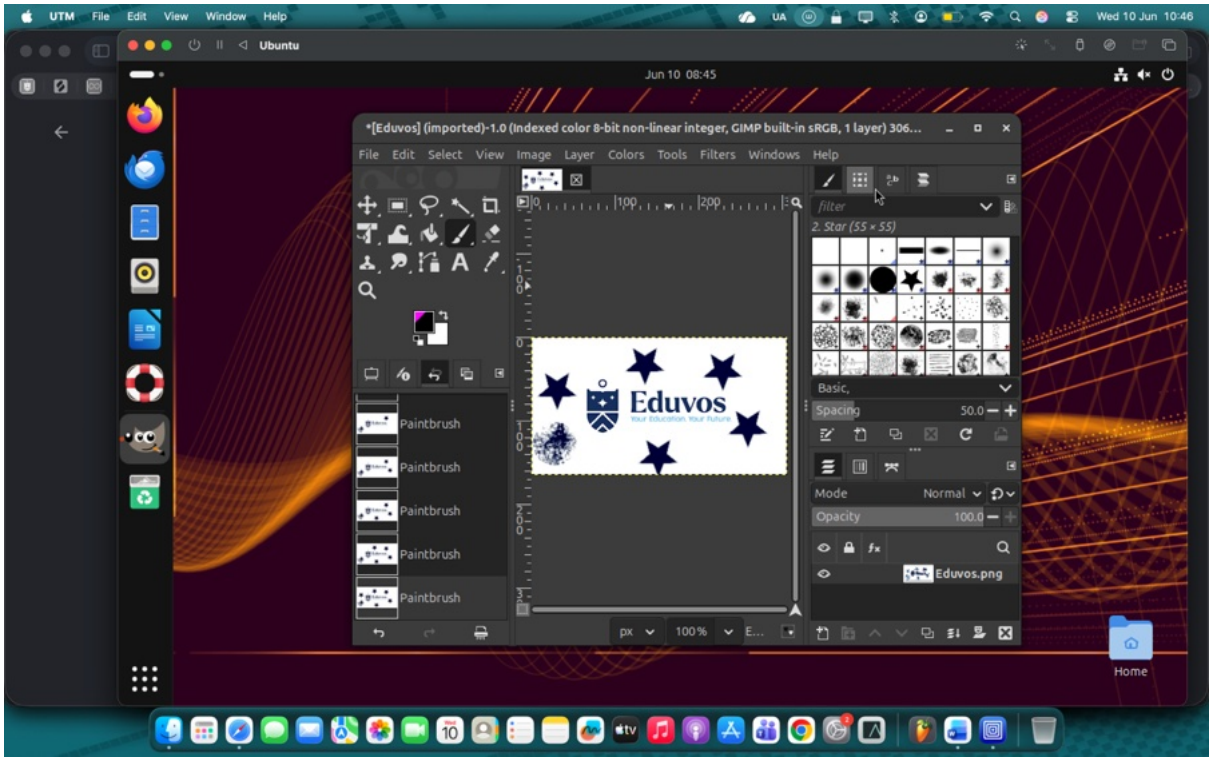

UTM File Edit View Window Help
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Ubuntu
Jun 10 08:25
Linux 2
QEMU 7.2 ARM
Linux
QEMU 7.2 ARM
Windows 2
QEMU 7.2 ARM
Windows 3
QEMU 7.2 ARM
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QEMU 7.2 ARM
Ubuntu
QEMU 7.2 ARM
LibreOffice
File Tools Help
Open File
Remote Files
Recent Documents
Templates
Create:
Writer Document
Calc Spreadsheet
Impress Presentation
Draw Drawing
Math Formula
Base Database
Filter: All Documents
Intern Guide
LibreOffice
Help
Donate
Home


UTM File Edit View Window Help
Wed 10 Jun 10:20
Ubuntu
Jun 10 08:20
Untitled 1 — LibreOffice Writer
File Edit View Insert Format Styles Table Form Tools Window Help
Default Paragraph Styl
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12 pt
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Name Intern Guide
Save
Home
Documents
Downloads
Music
Pictures
Videos
Other Locations
eduvovm
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Name
Size Type Modified
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Edit filter settings
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Page 1 of 1
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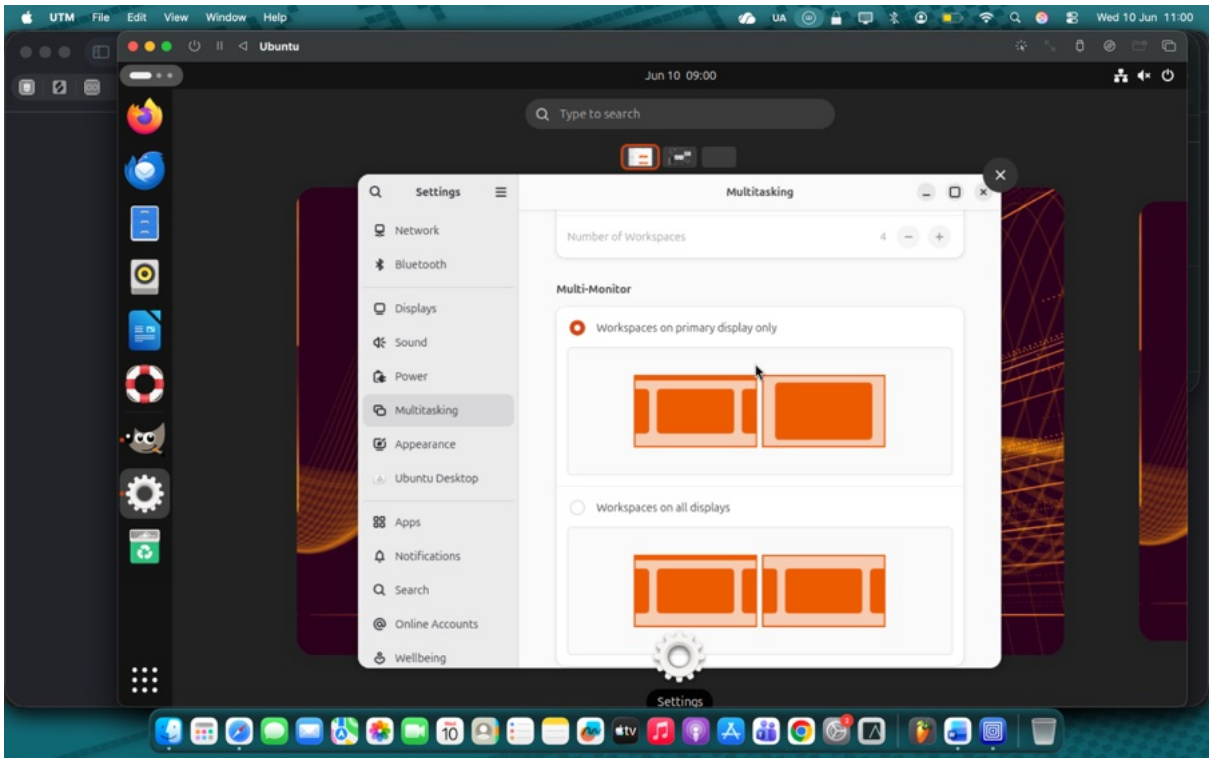


Creating and titling a document in LibreOffice with the name Intern Guide.

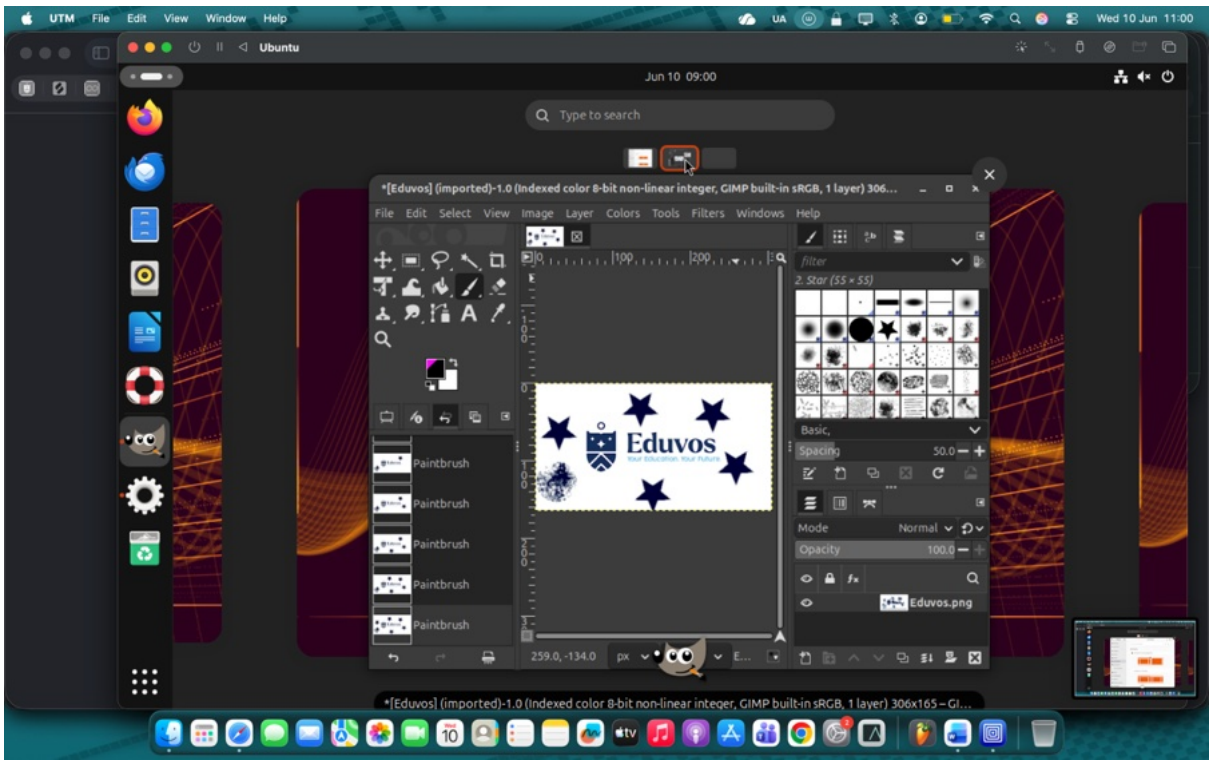




Using GIMP to edit an Image with Eduvos logo.



Using Workspaces to Organise and separate task System Setting from GIMP.



Using Workspaces to Organise and separate task GIMP from System Setting.

5) These application used support productivity

With FireFox providing distraction-reducing features like Reader View in a privacy focused web browser enabling quality online research.

With GIMP boosting creativity and documentation productivity by providing professional grade image editing and graphic design tools.

With LibreOffice providing a comprehensive office suite that increases workflow efficiency by offering Writer, Calc, and Impress in a free Microsft Office compatible environment for documentation creation and data processing.

And Workspaces which enhances multitasking by letting you separate different projects and groups of apps into distinct virtual desktop, eliminating clutter and improving your ability to focus.

6) Multitasking allows the operating system to run multiple applications and background processes at the same time improving CPU efficiency by sharing processing time among tasks.

Enhancing user productivity by letting users easily switch between programs without restarting.

7) Poor application management can lead to severe security vulnerabilities and system instability risks.

Where Failing to regularly update or patch installed applications creates exploitable security flaws and vulnerabilities.

And Installing software from unverified, third-party repositories or manually resolving conflicts can break application dependencies causing system instability.